LUMIR: an LLM-Driven Unified Agent Framework for Multi-task Infrared Spectroscopy Reasoning

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Abstract: Infrared spectroscopy enables rapid, non destructive analysis of chemical and material properties, yet high dimensional signals and overlapping bands hinder conventional chemometric methods. Large language models (LLMs), with strong generalization and reasoning capabilities, offer new opportunities for automated spectral interpretation, but their potential in this domain remains largely untapped. This study introduces LUMIR (LLM-driven Unified agent framework for Multi-task Infrared spectroscopy Reasoning), an agent based framework designed to achieve accurate infrared spectral analysis under low data conditions. LUMIR integrates a structured literature knowledge base, automated preprocessing, feature extraction, and predictive modeling into a unified pipeline. By mining peer reviewed spectroscopy studies, it identifies validated preprocessing and feature derivation strategies, transforms spectra into low dimensional representations, and applies few-shot prompts for classification, regression, and anomaly detection. The framework was validated on diverse datasets, including the publicly available Milk near-infrared dataset, Chinese medicinal herbs, Citri Reticulatae Pericarpium(CRP) with different storage durations, an industrial wastewater COD dataset, and two additional public benchmarks, Tecator and Corn. Across these tasks, LUMIR achieved performance comparable to or surpassing established machine learning and deep learning models, particularly in resource limited settings. This work demonstrates that combining structured literature guidance with few-shot learning enables robust, scalable, and automated spectral interpretation. LUMIR establishes a new paradigm for applying LLMs to infrared spectroscopy, offering high accuracy with minimal labeled data and broad applicability across scientific and industrial domains.

Keywords: Spectral analysis; Infrared spectroscopy; Large language models; In context learning; Automated chemometrics

1. Introduction

Infrared spectroscopy has emerged as a versatile analytical modality across food quality assessment, pharmaceutical process monitoring, agricultural product evaluation, and environmental sensing. Its non destructive nature, rapid data acquisition, and minimal sample preparation have rendered IR spectroscopy particular attraction for real time, in situ analysis in both academic research and industrial practice[1]. Nonetheless, the intrinsically high dimensionality of IR spectra, coupled with complex and overlapping absorption bands, imposes formidable challenges for robust, generalizable interpretation. Traditional machine learning and deep learning approaches ranging from partial least squares regression and support vector machines to convolutional neural networks and Transformer based architectures have achieved high accuracy on curated datasets through extensive feature engineering, model calibration, and hyperparameter optimization[2]. However, these methods often require large, representative training sets to mitigate overfitting, and they frequently necessitate retraining or recalibration when confronted with novel sample matrices or different instrumentation. Such dependencies introduce substantial costs in data collection and annotation and impede rapid deployment in variable real world environments.

Recent advances in large language models (LLMs) have demonstrated powerful capabilities in few-shot learning, chain of thought reasoning, and cross domain generalization[3][4]. Some pioneering studies have begun to apply LLMs to the scientific field, and LLMs have gradually developed into LLM based scientific agents that can automatically perform key scientific

LUMIR：一种用于多任务红外光谱推理的大语言模型驱动统一智能体框架

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摘要：红外光谱技术能够快速、无损地分析化学与材料特性，但高维信号和重叠谱带阻碍了传统化学计量学方法的应用。大语言模型（LLM）凭借强大的泛化与推理能力，为光谱自动解读提供了新机遇，然而其在该领域的潜力尚未得到充分挖掘。本研究提出了 LUMIR（基于大语言模型驱动的多任务红外光谱推理统一智能体框架），这是一种旨在低数据条件下实现精准红外光谱分析的智能体框架。LUMIR 将结构化文献知识库、自动化预处理、特征提取及预测建模整合为统一流程。通过挖掘同行评审的光谱学研究，该框架识别出经验证的预处理与特征衍生策略，将光谱转化为低维表示，并应用少样本提示进行分类、回归和异常检测。该框架在多个多样化数据集上进行了验证，包括公开的近红外牛奶数据集、中药材、不同贮藏年限的陈皮（CRP）、工业废水化学需氧量（COD）数据集，以及 Tecator 和 Corn 两个额外的公共基准数据集。在这些任务中，LUMIR 取得了媲美甚至超越既定机器学习和深度学习模型的性能，尤其在资源受限的场景下表现突出。本研究表明，结合结构化文献指导与少样本学习，可实现稳健、可扩展且自动化的光谱解读。LUMIR 确立了将大语言模型应用于红外光谱的新范式，仅需极少标注数据即可实现高精度，并在科学与工业领域具有广泛的适用性。**关键词：**光谱分析；红外光谱；大语言模型；上下文学习；自动化化学计量学

1. 引言

红外光谱法已成为一种多功能的分析模式，广泛应用于食品质量评估、制药过程监控、农产品评价以及环境传感领域。其无损特性、快速数据采集能力以及极少的样品前处理需求，使得红外光谱法在学术研究和工业实践中对实时原位分析具有特别的吸引力[1]。然而，红外光谱固有的高维特性，加之复杂且重叠的吸收带，给稳健且可泛化的解译带来了巨大挑战。传统的机器学习和深度学习方法——从偏最小二乘回归和支持向量机，到卷积神经网络和基于 Transformer 的架构——通过广泛的特征工程、模型校准和超参数优化，在精心构建的数据集上实现了高精度[2]。然而，这些方法通常需要大量具有代表性的训练集以减轻过拟合风险，并且在面对新型样品基质或不同仪器时，往往需要重新训练或重新校准。此类依赖性不仅引入了高昂的数据采集与标注成本，也阻碍了其在多变现实环境中的快速部署。

大型语言模型（LLM）的最新进展展示了其在少样本学习、思维链推理以及跨领域泛化方面的强大能力[3][4]。一些开创性研究已开始将 LLM 应用于科学领域，LLM 已逐渐发展为基于 LLM 的科学智能体，能够自动执行从假设生成、实验设计到数据分析和模拟等关键科学研究任务。

research tasks from hypothesis generation, experimental design to data analysis and simulation[5][6]. Unlike general purpose LLMs, these specialized agents integrate domain knowledge, advanced tool sets, and powerful verification mechanisms, enabling them to handle complex data types, ensure reproducible results, and promote scientific breakthroughs. LLMs are becoming a general tool for connecting computational methods, experimental data, literature text resources, and domain expertise[7]. In recent work on spectral analysis and detection, large language models (LLMs) have been applied in two complementary ways. One approach treats an LLM as an interpreter: high dimensional spectral signals are preprocessed and serialized (for example into JSON), then presented to the model through task specific prompt engineering or in context examples so the model can extract features, highlight diagnostic bands, or produce quantitative estimates without updating model weights. A second approach finetunes an LLM (or LLM derived architecture) on formatted spectral sequences so the model directly maps spectra to analytical targets. For example, Zhu et al. (2024) finetuned a Babbage-002 model on subpixel decomposed NIR spectra encoded in JSON to predict the toxin content, reporting higher accuracy than 1-D and 2-D CNN baselines[8]. Liang et al. (2024) used a prompt driven LLM workflow to extract characteristic UV-NIR spectral bands for chemical oxygen demand prediction, showing that carefully designed prompts and in context reasoning can match or exceed traditional machine learning methods while substantially reducing the need for task specific parameter tuning[9].

In parallel, several related end-to-end frameworks have emerged in other scientific domains. For instance, Large Language Model-Informed X-ray Photoelectron Spectroscopy (XPS) analysis integrates GPT-based reasoning with curve-fitting procedures to assist in interpreting photoelectron spectra[10], but it focuses primarily on post-processing rather than building a generalized multi-task pipeline. SpecCLIP introduces a foundation model that aligns and translates stellar spectroscopic measurements across different astronomical instruments[11], emphasizing cross-instrument calibration instead of chemometric method selection. More recently, XASDAML delivers a machine-learning-based platform for X-ray absorption spectroscopy, covering dataset construction, modeling, and statistical analysis[12], though it relies on conventional ML workflows rather than LLM-driven agents. Compared with these works, our framework uniquely integrates literature-guided preprocessing, feature extraction, and multi-task reasoning into a single LLM-driven agent, enabling automated spectral analysis across diverse infrared datasets with enhanced adaptability and interpretability.

To address the limitations of traditional chemometrics, we propose a unified, LLM-driven agent framework that seamlessly integrates literature based method retrieval, automated spectral preprocessing, feature extraction, and multi-task reasoning within a single end-to-end pipeline. A structured knowledge base of peer reviewed IR spectroscopy publications is first queried via retrieval algorithms to identify relevant preprocessing and feature extraction methods for the specified research object. These method descriptions are then mapped to a predefined Python function librarycomprising Asymmetric Least Squares baseline correction, Savitzky-Golay smoothing, standard normal variate transformation, multiplicative scatter correction, normalization, detrending, principal component analysis (PCA)[13], partial least squares (PLS) regression[14], Nonnegative Matrix Factorization (NMF)[15], continuous wavelet transform[16], spectral derivatives, peak detection, and statistical summarization without exposing raw high dimensional spectral arrays to the LLM. The agent invokes the selected functions in sequence to transform the spectral data into structured feature sets, ensuring that each step remains scientifically grounded and reproducible. Figure 1 shows the basic structure of an LLM-Driven Unified Agent Framework for Multi-task Infrared Spectroscopy Reasoning.

与通用 LLM 不同，这些专用智能体集成了领域知识、先进工具集和强大的验证机制，使其能够处理复杂数据类型、确保结果可复现并推动科学突破。LLM 正成为连接计算方法、实验数据、文献文本资源和领域专业知识的通用工具[7]。在近期的光谱分析与检测研究中，大型语言模型（LLM）以两种互补的方式得到应用。一种方法将 LLM 视为解释器：高维光谱信号经过预处理和序列化（例如转换为 JSON），然后通过特定任务的提示工程或上下文示例呈现给模型，使模型能够在不更新权重的情况下提取特征、突出诊断波段或生成定量估计。另一种方法则在格式化的光谱序列上对 LLM（或源自 LLM 的架构）进行微调，使模型能够直接将光谱映射到分析目标。例如，Zhu 等人（2024）在编码为 JSON 的子像素分解近红外光谱上微调了 Babbage-002 模型以预测毒素含量，报告其准确率高于 1D 和 2D CNN 基线[8]。Liang 等人（2024）利用提示驱动的 LLM 工作流提取用于化学需氧量预测的特征紫外 - 近红外光谱波段，表明精心设计的提示和上下文推理可以达到甚至超越传统机器学习方法的性能，同时大幅减少对特定任务参数调整的需求[9]。

与此同时，其他科学领域也涌现出若干相关的端到端框架。例如，大语言模型赋能的 X 射线光电子能谱（XPS）分析将基于 GPT 的推理与曲线拟合程序相结合，以辅助解读光电子能谱[10], ，但其主要侧重于后处理，而非构建通用的多任务流程。SpecCLIP 引入了一种基础模型，用于对齐并转换不同天文仪器间的恒星光谱测量数据[11], ，其重点在于跨仪器校准，而非化学计量方法的选择。最近，XASDAML 提供了一个基于机器学习的 X 射线吸收光谱平台，涵盖数据集构建、建模和统计分析[12], ，但它依赖的是传统机器学习工作流，而非由大语言模型驱动的智能体。与这些工作相比，我们的框架独特地将文献引导的预处理、特征提取和多任务推理集成到单个大语言模型驱动的智能体中，从而能够在多样化的红外数据集上实现自动化光谱分析，并具备更强的适应性和可解释性。

为解决传统化学计量学的局限性，我们提出了一种统一的、由大语言模型（LLM）驱动的智能体框架。该框架在一个端到端的流程中无缝集成了基于文献的方法检索、自动光谱预处理、特征提取和多任务推理。首先，通过检索算法查询包含同行评审红外光谱出版物的结构化知识库，以识别针对特定研究对象的相關预处理和特征提取方法。随后，将这些方法描述映射到一个预定义的 Python 函数库，该库包含非对称最小二乘基线校正、Savitzky-Golay 平滑、标准正态变量变换、多元散射校正、归一化、去趋势、主成分分析（PCA）[13],偏最小二乘（PLS）回归[14], 非负矩阵分解（NMF）[15], 连续小波变换[16], 光谱导数、峰检测及统计汇总，且无需向 LLM 暴露原始高维光谱数组。智能体按顺序调用所选函数，将光谱数据转换为结构化特征集，确保每一步都具有科学依据且可复现。图 1 展示了用于多任务红外光谱推理的 LLM 驱动统一智能体框架的基本结构。

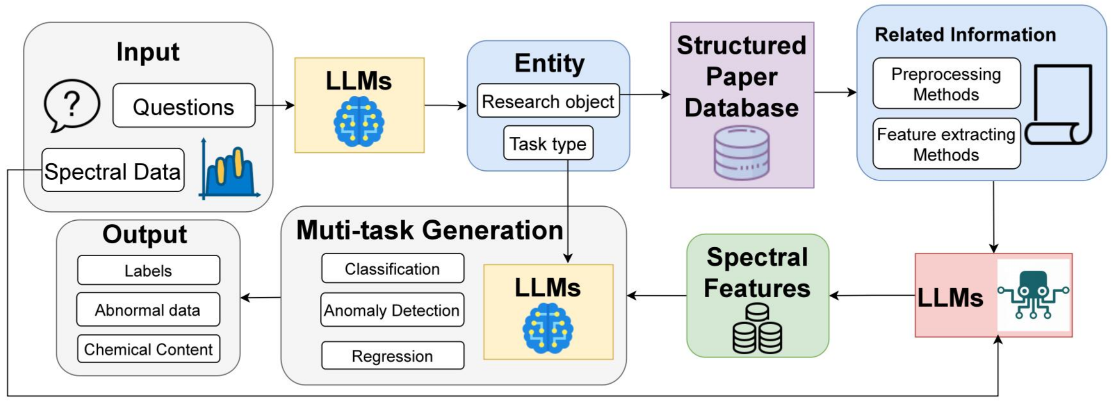


Figure 1. Basic structure of an LLM-Driven Unified Agent Framework for Multi-task Infrared Spectroscopy Reasoning

Building upon this automated preprocessing and feature extraction backbone, the framework integrates multi-task reasoning capabilities that allow the LLM to perform downstream analytical tasks such as classification, regression, and anomaly detection within a unified setting. For each task, the model is provided with few-shot examples drawn from the processed feature sets, and predictions are generated directly based on these contextual demonstrations. By embedding structured guidance and task specific prompting into the workflow, the proposed framework automates the selection and execution of chemometric methods and adapts flexibly to diverse analytical objectives, thereby offering an interpretable and scalable solution for infrared spectral analysis.

The key contributions of this work are:

1. End-to-end intelligent spectral analysis pipeline. We integrate a priori literature knowledge to select scientifically validated preprocessing and feature-extraction routines, then apply multi-turn LLM inference to execute diverse analytical tasks within a single workflow.

2. Unified multi-task framework. Our system concurrently performs classification, regression, and anomaly detection, overcoming the constraints of single-objective methods.

3. Few-shot prompting efficiency. By leveraging only minimal training data, the agent achieves high accuracy that consistently surpasses traditional machine learning models trained on equivalent sample sizes.

2. Materials and Method

2.1 Experimental materials

This study addressed three spectroscopic tasks, namely classification, anomaly detection, and regression, each of which was associated with specific sample sets and corresponding wavelength ranges. In the classification and anomaly detection experiments, three categories of materials were investigated. The classification task was designed to distinguish subclasses within a broader material category by assigning each spectrum to a specific subclass label, while the anomaly detection task was formulated as a binary problem to determine whether a sample belonged to a predefined subclass (True) or not (False).

The first dataset comprised milk samples with varying concentrations of lactose, measured within the 1100–2300 nm spectral range. This Milk NIR Spectral Dataset is an openly available benchmark provided by the NIRPyResearch initiative, intended to demonstrate the use of near-infrared spectroscopy for discriminating lactose content[17]. It includes 450 spectral scans obtained from nine distinct formulations, each representing a different mixture ratio of regular and lactose free milk. The second dataset contained three types of Chinese medicinal materials, namely Lonicerae Japonicae Flos, Lonicerae Flos, and their mixture, analyzed in the 900–1700 nm spectral range. The third dataset included Citri Reticulatae Pericarpium samples with storage durations ranging from one to eight years, with spectra also recorded between 900 and 1700 nm.

For the regression task, an internal wastewater dataset was used with the objective of predicting the chemical oxygen demand (COD) concentration of influent samples collected prior to treatment at a municipal wastewater treatment plant in Chongqing, China. In addition to this industrial dataset, two widely used public datasets, Tecator and Corn, were incorporated.

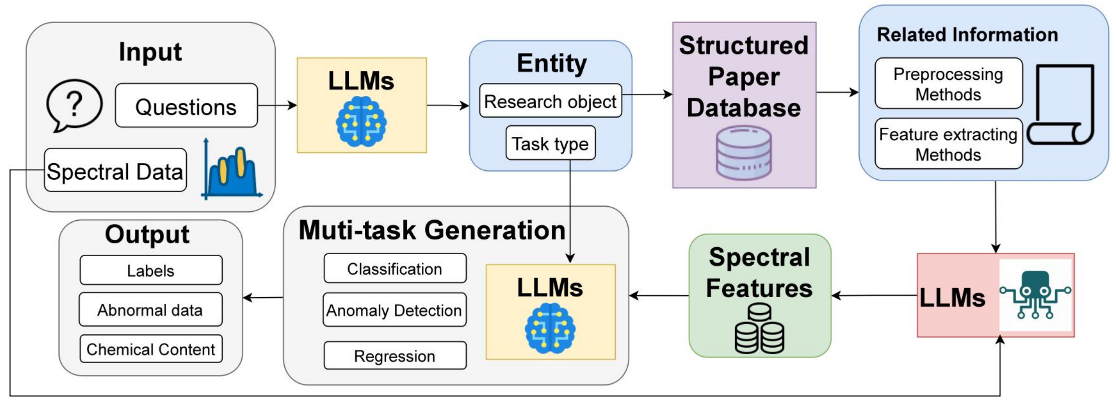


图 1. 用于多任务红外光谱推理的大语言模型驱动统一智能体框架的基本结构

在此自动预处理和特征提取骨干的基础上，该框架集成了多任务推理能力，使 LLM 能够在统一设置中执行分类、回归和异常检测等下游分析任务。对于每项任务，模型会获得来自处理后特征集的少样本示例，并直接基于这些上下文演示生成预测。通过将结构化指导和特定任务的提示嵌入工作流，所提出的框架实现了化学计量学方法的选择与执行自动化，并能灵活适应多样的分析目标，从而为红外光谱分析提供了一种可解释且可扩展的解决方案。

本工作的主要贡献如下：

1. 端到端智能光谱分析流程。我们整合先验文献知识，以选择经科学验证的预处理和特征提取程序，随后应用多轮大语言模型推理，在单一工作流中执行多样化的分析任务。

2. 统一多任务框架。我们的系统可并发执行分类、回归和异常检测任务，克服了单目标方法的局限性。

3. 少样本提示效率。通过仅利用极少量的训练数据，该智能体实现了高准确率，其表现始终优于在同等样本量下训练的传统机器学习模型。

2. 材料与方法

2.1 实验材料

本研究涵盖了三项光谱任务，即分类、异常检测和回归，每项任务均关联特定的样本集及相应的波长范围。在分类与异常检测实验中，研究了三类材料。分类任务旨在通过将每个光谱分配至特定的子类标签，以区分更广泛材料类别内的子类；而异常检测任务则被构建为一个二分类问题，用于判定样本是否属于预定义的子类（真）或不属于（假）。

第一个数据集包含乳糖浓度各异的牛奶样本，测量光谱范围为 1100–2300 nm。该牛奶近红外光谱数据集是由 NIRPyResearch 项目提供的公开基准数据集，旨在展示利用近红外光谱技术鉴别乳糖含量的方法[17]。该数据集包含从九种不同配方中获取的 450 次光谱扫描，每种配方代表普通牛奶与无乳糖牛奶的不同混合比例。第二个数据集包含三种中药材，即金银花、山银花及其混合物，分析光谱范围为 900–1700 nm。第三个数据集包含储存年限从一年至八年不等的陈皮样本，其光谱同样记录于 900 至 1700 nm 之间。

针对回归任务，本研究使用了一个内部废水数据集，旨在预测中国重庆某市政污水处理厂在处理前收集的进水样本的化学需氧量（COD）浓度。除了该工业数据集外，还引入了两个广泛使用的公共数据集：Tecator 和 Corn。

The Tecator dataset consists of absorbance spectra of chopped meat accompanied by reference measurements of fat, water, and protein, and the canonical regression problem focuses on predicting fat content[18]. The Corn dataset, made available by Eigenvector Research, contains eighty samples measured using three near-infrared spectrometers covering the 1100–2498 nm range with two nanometre spacing, together with reference values for moisture, oil, protein, and starch[19]. The main purpose is to predict the protein content in corn in this study.

The integration of internal and public datasets ensured complementary coverage of application scenarios and spectral ranges. The internal datasets enabled targeted evaluation of classification and anomaly detection in food and traditional medicine, and supported a realistic regression task in wastewater analysis. Public benchmarks, by contrast, offered well documented provenance, stable sampling grids, and standardized reference targets, thereby enhancing reproducibility and enabling direct comparison with conventional machine learning and deep learning baselines under consistent preprocessing and evaluation protocols.

Table 1 Experimental material data summary

Sample Name	Samples	Wavelength Range	Task Type	Note
Milk	360	1100-2300nm	Classification, Anomaly Detection	9 classes with different mixture ratios of regular vs lactose free milk
Chinese medicinal herbs	120	900-1700nm	Classification, Anomaly Detection	3 classes: Lonicerae Japonicae Flos (LJF), Lonicerae Flos (LF) and the mixture of them
Citri Reticulatae Pericarpium	240	900-1700nm	Classification, Anomaly Detection	8 classes with storage durations ranging from 1 to 8 years
Wastewater	67	190-1100nm	Regression	Samples with different COD
Tecator	215	850-1050 nm	Regression	Samples with different fat content
Corn	80	1100-2498 nm	Regression	Samples with different protein content

2.2 Structured Knowledge Base and Entity Retrieval

In order to endow our automated spectral analysis framework with domain specific knowledge, we built upon the Spectral Data Analysis and Application Platform (SDAAP) dataset developed in our prior work, which represents the first systematically curated, open access text corpus dedicated to spectroscopic analysis and detection[20]. SDAAP not only aggregates and annotates peer reviewed publications across diverse spectroscopic applications, but also supplies each record with structured "knowledge guidance" entries that specify experimental targets, spectral modalities, preprocessing protocols, feature extraction techniques, and modeling approaches. This resource addresses a critical deficiency in publicly available text datasets for spectroscopy and provides a solid foundation for subsequent LLM applications in this domain.

From the SDAAP repository, we filtered entries that were directly relevant to the materials investigated in this study, including milk dataset, traditional Chinese medicinal herbs, the Tecator and Corn datasets, and Citri Reticulatae Pericarpium, with the aim of aligning the literature knowledge base with the experimental datasets described in Section 2.1. Each selected publication was processed using MinerU, an open source automated literature mining tool[21], and the extracted annotations were subsequently integrated into a large language model pipeline to obtain structured metadata. The metadata encompassed the material or analyte under investigation, the spectroscopic technique and wavelength ranges employed, the optimal preprocessing workflow applied to the raw spectra, the most effective feature derivation strategy, and the predictive modeling approach adopted in each study.

Within our end-to-end spectral analysis system, user input comprises two modalities: a natural language query and raw spectral measurements. Upon receiving a query, the system invokes an LLM with a bespoke prompt engineered for entity recognition. The LLM parses the question to identify two key entities: the research object and the analysis task type

Tecator 数据集包含碎肉的吸收光谱以及脂肪、水分和蛋白质的参考测量值，其典型的回归问题侧重于预测脂肪含量[18]。Corn 数据集由 Eigenvector Research 提供，包含 80 个样本，这些样本使用三台近红外光谱仪在 1100–2498 nm 范围内以 2 纳米间隔进行测量，并配有水分、油脂、蛋白质和淀粉的参考值[19]。本研究的主要目的是预测玉米中的蛋白质含量。

内部数据集与公共数据集的整合确保了应用场景和光谱范围的互补覆盖。内部数据集支持对食品和传统医药领域的分类与异常检测进行针对性评估，并为废水分析中的回归任务提供了现实依据。相比之下，公共基准数据集具有来源记录详尽、采样网格稳定以及参考目标标准化等特点，从而提升了可复现性，并使得在一致的数据预处理和评估协议下，能够直接与传统的机器学习和深度学习基线方法进行对比。

表 1 实验材料数据汇总

样本名称	样本	波长范围	任务类型	Note
Milk	360	1100-2300nm	分类、异常检测	9 个类别，包含不同混合比例的普通牛奶与无乳糖牛奶
中草药	120	900-1700nm	分类、异常检测	3 个类别：金银花 (LJF)、山银花 (LF) 及其混合物的混合物
柑橘果皮	240	900-1700nm	分类、异常检测	8 个类别，存储时长为 1 至 8 年不等
废水	67	190-1100nm	回归	具有不同 COD 的样品
Tecator	215	850-1050 nm	回归	具有不同脂肪含量的样品
Corn	80	1100-2498 nm	回归	具有不同蛋白质含量的样品

2.2 结构化知识库与实体检索

为了赋予我们的自动光谱分析框架领域特定知识，我们基于先前工作中开发的光谱数据分析与应用平台（SDAAP）数据集构建了该框架，这是首个专门用于光谱分析与检测的系统性整理、开放获取文本语料库[20]。SDAAP 不仅汇总并标注了涵盖多种光谱应用的同行评审出版物，还为每条记录提供了结构化的“知识指导”条目，具体指明实验目标、光谱模态、预处理方案、特征提取技术及建模方法。这一资源弥补了 publicly available 光谱文本数据集的关键缺失，并为该领域后续的大语言模型应用奠定了坚实基础。

从 SDAAP 存储库中，我们筛选出与本研究所调查材料直接相关的条目，包括牛奶数据集、传统中草药、Tecator 和 Corn 数据集以及柑橘皮，旨在使文献知识库与 2.1 节中描述的实验数据集保持一致。每篇选定的出版物均使用 MinerU（一种开源自动化文献挖掘工具[21], ）进行处理，随后将提取的注释整合到大语言模型流水线中以获取结构化元数据。这些元数据涵盖了各研究中调查的材料或分析物、所采用的光谱技术及波长范围、应用于原始光谱的最佳预处理工作流程、最有效的特征推导策略以及所采用的预测建模方法。

在我们的端到端光谱分析系统中，用户输入包含两种模态：自然语言查询和原始光谱测量数据。收到查询后，系统会调用大语言模型（LLM），并使用专为实体识别设计的定制提示词。该大语言模型解析问题以识别两个关键实体：研究对象和分析任务类型。

(classification, anomaly detection, or regression). Once the research object entity is identified, retrieval algorithms are used to search the structured knowledge base for records whose material descriptors match or closely resemble the query entity. The retrieved structured entries are then forwarded, alongside the raw spectral data, to a downstream LLM agent module that synthesizes prior knowledge with the new measurements. Concurrently, the extracted task type entity informs a multi-task inference controller within the LLM agent, directing it to apply the appropriate reasoning paradigm whether discriminative classification, binary anomaly detection, or continuous value regression to produce the final analytical output. Figure 2 shows the specific process of entity extraction and knowledge retrieval.

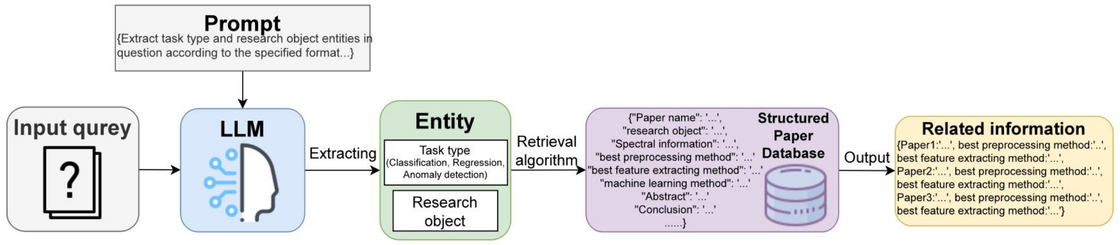


Figure 2. The specific process of entity extraction and knowledge retrieval

2.3 LLM Agent for Spectral Preprocessing and Feature Extraction

The LLM agent drives spectral data conditioning and representation by translating structured metadata from the knowledge base into a methodology selection framework. By default, upon receiving a raw absorbance spectrum and the extracted entity, the agent evaluates the top-k most relevant literature sources and determines the preprocessing and feature extraction pipeline through majority voting over these references. In cases where multiple pipelines appear with equal frequency, the system defaults to the first valid pipeline encountered in the ranked list. The selected algorithms from the spectral analysis library are then applied, outputs are validated against predefined quality assurance metrics, and, if necessary, the process is iterated until satisfactory spectral quality is achieved.

To provide transparency and user control, the interface displays the top-k relevant literature sources along with the methods reported in them. If the user is not satisfied with the automatically chosen pipeline, they can directly override the selection by specifying any one of the listed literature sources, in which case the agent immediately re-executes the processing with the corresponding method pair.

Baseline correction(BC) is offered via the asymmetric least squares (AsLS) algorithm, which estimates the corrected spectrum z by minimizing[22].

$$\sum_{i=1}^{n}(y_i - z_i)^2 + \lambda \sum_{i=2}^{n}(z_{i+1} - 2z_i + z_{i-1})^2 , \tag{1}$$

where y_i denotes the raw intensity at index i, $z_i$the baseline subtracted intensity, and λ a smoothing penalty that balances fidelity to the original spectrum against baseline smoothness.

Spectral smoothing is performed with the Savitzky-Golay(SG) filter, which fits a polynomial of degree over a moving window of width $2m + 1$ and computes the smoothed value[23].

$$\hat{y}_i = \sum_{j=-m}^{m} c_j\, y_{i+j} \tag{2}$$

where the convolution coefficients c_j arise from the least squares fit of the local polynomial.

Normalization options include min-max scaling, which linearly rescales the intensities as

$$y_i^{'} = \frac{y_i - \min_k y_k}{\max_k y_k - \min_k y_k} \tag{3}$$

and the standard normal variate (SNV) transform[24], which centers and scales each spectrum by its own statistics:

$$y_i^{''} = \frac{y_i - \bar{y}}{\sigma_y} \tag{4}$$

（分类、异常检测或回归）。一旦识别出研究对象实体，检索算法便会在结构化知识库中搜索材料描述符与查询实体匹配或高度相似的记录。随后，这些检索到的结构化条目将与原始光谱数据一同发送至下游的大语言模型代理模块，该模块负责将先验知识与新测量数据进行综合。与此同时，提取出的任务类型实体会告知大语言模型代理内的多任务推理控制器，指导其应用适当的推理范式——无论是判别式分类、二值异常检测还是连续值回归——以生成最终的分析输出。图 2 展示了实体提取与知识检索的具体流程。

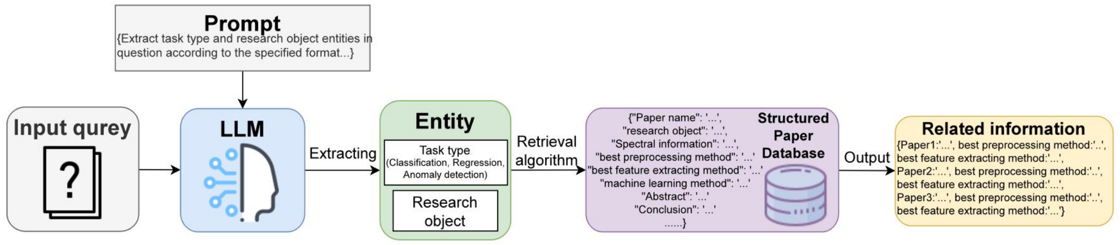


图 2. 实体提取与知识检索的具体流程

2.3 用于光谱预处理和特征提取的大语言模型智能体

大语言模型智能体通过将知识库中的结构化元数据转化为方法选择框架，驱动光谱数据的调理与表征。默认情况下，在接收到原始吸收光谱和提取的实体后，智能体会评估最相关的前 k 篇文献来源，并通过对这些参考文献进行多数投票来确定预处理和特征提取流程。若出现多个流程频率相同的情况，系统则默认采用排序列表中遇到的第一个有效流程。随后应用从光谱分析库中选定的算法，针对预定义的质量保证指标对输出进行验证；如有必要，将迭代该过程直至获得满意的光谱质量。

为提供透明度和用户控制权，界面会显示最相关的前 k 篇文献来源及其所报告的方法。如果用户对自动选择的流程不满意，可以直接通过指定所列文献中的任意一篇来覆盖该选择；此时，智能体将立即使用相应的方法对重新执行处理。

基线校正（BC）通过非对称最小二乘（AsLS）算法实现，该算法通过最小化[22]来估算校正后的光谱。

$$\sum_{i=1}^{n}(y_i - z_i)^2 + \lambda \sum_{i=2}^{n}(z_{i+1} - 2z_i + z_{i-1})^2 , \tag{1}$$

其中，y_i 表示索引 i 处的原始强度，z_i表示扣除基线后的强度，而 λ 是一个平滑惩罚项，用于平衡对原始光谱的拟合度与基线的平滑度。

光谱平滑采用 Savitzky-Golay（SG）滤波器执行，该滤波器在宽度为 2m+ 1 的移动窗口内拟合一个多项式，并计算平滑值[23]。

$$\hat{y}_i = \sum_{j=-m}^{m} c_j\, y_{i+j} \tag{2}$$

其中，卷积系数$_j$ 源于局部多项式的最小二乘拟合。

归一化选项包括最小 - 最大缩放，该方法对强度进行线性重缩放，公式如下：

$$y_i^{'} = \frac{y_i - \min_k y_k}{\max_k y_k - \min_k y_k} \tag{3}$$

以及标准正态变量（SNV）变换[24], ，该变换利用光谱自身的统计量对其进行中心化和缩放：

$$y_i^{''} = \frac{y_i - \bar{y}}{\sigma_y} \tag{4}$$

with $\bar{y}$ and σ_y representing the mean and standard deviation of the spectrum, respectively.

When derivative spectra are required, the first derivative is approximated by

$$y_i^{(1)} = \frac{y_{i+1} - y_{i-1}}{2\Delta\lambda} \tag{5}$$

and the second derivative by

$$y_i^{(2)} = \frac{y_{i+1} - 2y_i + y_{i-1}}{\Delta\lambda^2} \tag{6}$$

where $\Delta\lambda$ is the uniform wavelength increment.

After preprocessing, the agent proceeds to feature extraction. Options include principal component analysis (PCA), partial least squares (PLS) regression, continuous wavelet transform (CWT), Nonnegative Matrix Factorization (NMF) and a Lambert-Beer-Pearson correlation approach. In the fully automated workflow, the method most relevant to the input task type is executed; however, the user can override this choice via the same clickable method list shown in the interface. PCA is implemented by solving the eigenvalue problem.

$$Cu_k = \lambda_k u_k, \quad C = \frac{1}{n-1} X^\top X \tag{7}$$

where X is the preprocessed data matrix, u_k are the principal directions, and λ_k their variances. PLS iteratively derives latent scores by maximizing covariance between the predictor matrix X and response vector y, extracting weight vectors w and loadings p, q through

$$\max_{\|w\|=1} Cov(Xw, y), \quad X \leftarrow X - tp^\top, \quad y \leftarrow y - tc \tag{8}$$

where $t = $ Xw, $u = $ yq, and c is the loading on y. The CWT is computed as

$$W(a,b) = \frac{1}{\sqrt{a}} \int_{-\infty}^{\infty} y(\lambda)\psi\left(\frac{\lambda - b}{a}\right) d\lambda \tag{9}$$

with scale a, translation b, and mother wavelet ψ.

In scenarios where a small set of reference spectra with known analyte concentrations is available,such as calibration of chemical oxygen demand or quantification of specific compound concentrations,the agent can instead invoke a hybrid Lambert-Beer-Pearson method[25]. Here, the absorbance at each wavelength λ is first computed via the Lambert-Beer law,

$$A(\lambda) = \log \frac{I_0(\lambda)}{I(\lambda)} \tag{10}$$

where I_0 is the incident light intensity and I the transmitted intensity. These absorbance spectra are then compared against each reference curve $R_{ij}(\lambda)$ by computing the Pearson correlation coefficient

$$r_j = \frac{\sum_i (A_i - \bar{A})(R_{ij} - \bar{R}_j)}{\sqrt{\sum_i (A_i - \bar{A})^2} \sqrt{\sum_i (R_{ij} - \bar{R}_j)^2}} \tag{11}$$

The top n coefficients $\{r_j\}$ serve as features that capture both the linear absorbance behavior and the spectral similarity to calibration standards.

This design ensures that the pipeline is fully automatic by default while allowing rapid, transparent user overrides when domain expertise suggests a different approach. Figure 3 shows the specific process of preprocessing and feature extraction.

其中 y 和 σ_y 分别代表光谱的均值和标准差。

当需要导数光谱时，一阶导数可通过以下公式近似：

$$y_i^{(1)} = \frac{y_{i+1} - y_{i-1}}{2\Delta\lambda} \tag{5}$$

二阶导数则通过以下公式近似：

$$y_i^{(2)} = \frac{y_{i+1} - 2y_i + y_{i-1}}{\Delta\lambda^2} \tag{6}$$

其中 $\Delta\lambda$ 为均匀的波长增量。

预处理完成后，代理将进入特征提取阶段。可选方法包括主成分分析（PCA）、偏最小二乘（PLS）回归、连续小波变换（CWT）、非负矩阵分解（NMF）以及 Lambert-Beer-Pearson 相关性方法。在全自动工作流程中，系统将执行与输入任务类型最相关的方法；但用户也可以通过界面中显示的相同可点击方法列表来覆盖此选择。PCA 通过求解特征值问题来实现。

$$Cu_k = \lambda_k u_k, \quad C = \frac{1}{n-1} X^\top X \tag{7}$$

其中 X 为预处理后的数据矩阵，$_k$ 为主成分方向，$_k$ 为其方差。PLS 通过最大化预测矩阵 X 与响应向量 y 之间的协方差，迭代推导潜在得分，并通过以下公式提取权重向量 w 以及载荷p、q：

$$\max_{\|w\|=1} Cov(Xw, y), \quad X \leftarrow X - tp^\top, \quad y \leftarrow y - tc \tag{8}$$

其中 = Xw、 = yq，且 c 为 y 上的载荷。CWT 的计算公式如下：

$$W(a,b) = \frac{1}{\sqrt{a}} \int_{-\infty}^{\infty} y(\lambda)\psi\left(\frac{\lambda - b}{a}\right) d\lambda \tag{9}$$

包括尺度、平移和母小波。

在仅有少量已知分析物浓度的参考光谱可用的场景中（例如化学需氧量的校准或特定化合物浓度的定量），代理可以转而调用混合朗伯 - 比尔 - 皮尔逊方法[25]。在此方法中，首先通过朗伯 - 比尔定律计算每个波长处的吸光度，

$$A(\lambda) = \log \frac{I_0(\lambda)}{I(\lambda)} \tag{10}$$

其中 I_0 是入射光强度，I 是透射光强度。随后，通过计算皮尔逊相关系数，将这些吸光度光谱与每条参考曲线 $R_{ij}(\lambda)$ 进行比较。

$$r_j = \frac{\sum_i (A_i - \bar{A})(R_{ij} - \bar{R}_j)}{\sqrt{\sum_i (A_i - \bar{A})^2} \sqrt{\sum_i (R_{ij} - \bar{R}_j)^2}} \tag{11}$$

前 n 个系数{rj}作为特征，既捕捉了线性吸光度行为，也反映了与校准标准的光谱相似性。

该设计确保流水线默认完全自动化，同时当领域专业知识表明需要不同方法时，允许用户进行快速、透明的干预。图 3 展示了预处理和特征提取的具体流程。

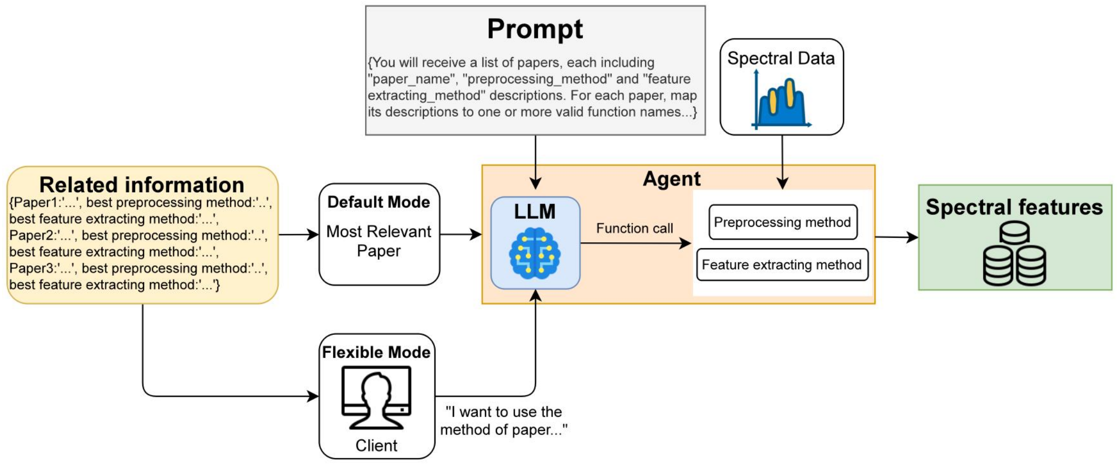


Figure 3. The specific process of preprocessing and feature extracting

2.4 LLM Multi-task Reasoning Pipeline

Classification, anomaly detection, and regression are performed by a large language model with fixed parameters, and behavior is governed by the prompt rather than by weight updates. For each dataset, a compact set of exemplar pairs from the training partition is assembled into an instruction oriented prompt that specifies task definition, spectral preprocessing choices, feature representations, and a small number of labeled exemplars. Given a query spectrum x_i, the model produces a categorical label for classification and anomaly detection or a scalar response for regression. No gradient based optimization is involved. All adaptation arises from information encoded in the prompt and from the curated exemplars that anchor task specific reasoning. Figure 4 illustrates the overall workflow.

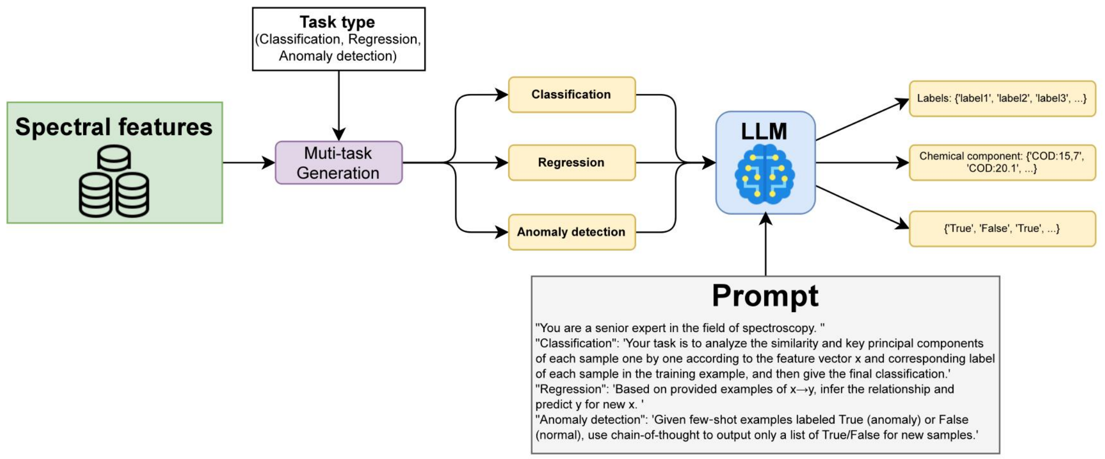


Figure 4. The specific process of LLM multi-task reasoning

Performance is evaluated on the held out test partition using task appropriate metrics. Let T denote the test set and $N_T = |T|$. For classification, accuracy is computed as

$$\mathrm{Acc} = \frac{1}{N_T}\sum_{i\in T} 1\,[\hat{y}_i = y_i] \tag{12}$$

For the anomaly detection task, the area under the receiver operating characteristic curve (AUC) is employed, defined as

$$\mathrm{AUC}^{(t)} = \frac{1}{N_{pos}N_{neg}}\sum_{i:y_i=1}\sum_{j:y_j=0}\mathbf{1}\left(s_i^{(t)} > s_j^{(t)}\right) \tag{13}$$

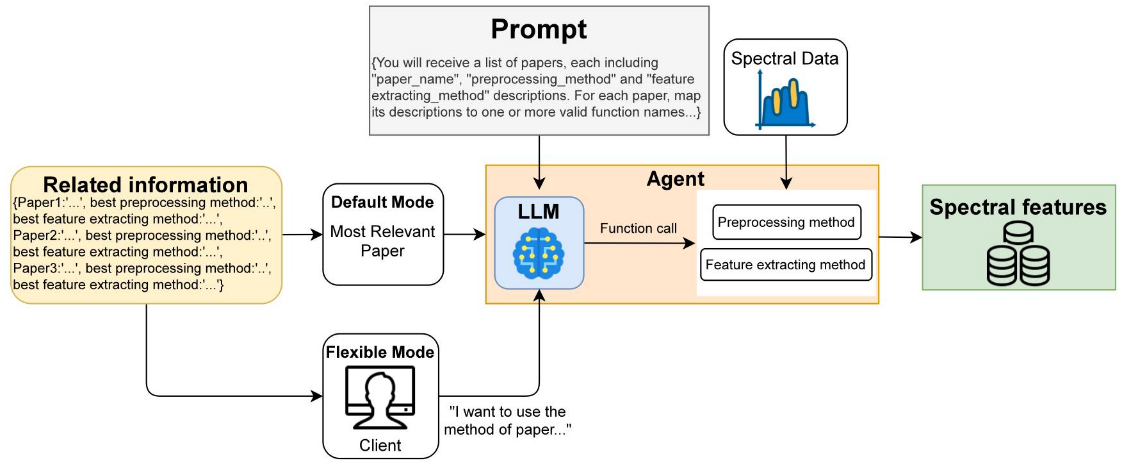


图 3. 预处理与特征提取的具体流程

2.4 大语言模型多任务推理流水线

分类、异常检测和回归由参数固定的大语言模型执行，其行为由提示词（prompt）而非权重更新来调控。对于每个数据集，从训练分区中选取一组紧凑的示例对，构建成面向指令的提示词，其中明确任务定义、光谱预处理选项、特征表示形式以及少量带标签的示例。给定查询光谱 x_i，模型会输出用于分类和异常检测的类别标签，或用于回归的标量响应。整个过程不涉及基于梯度的优化，所有适配均源于提示词中编码的信息以及锚定特定任务推理的精选示例。图 4 展示了整体工作流程。

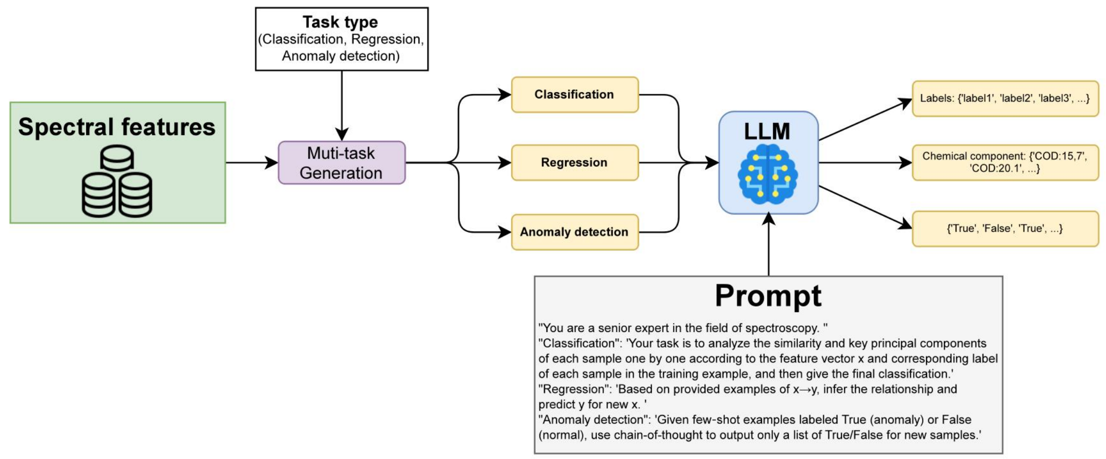


图 4. 大语言模型多任务推理的具体流程

性能评估在保留的测试集上进行，采用适用于各任务的指标。设 T 表示测试集，N_T =T。对于分类任务，准确率|计算如下:

$$\mathrm{Acc} = \frac{1}{N_T}\sum_{i\in T} 1\,[\hat{y}_i = y_i] \tag{12}$$

对于异常检测任务，采用受试者工作特征曲线（AUC）下的面积，其定义为

$$\mathrm{AUC}^{(t)} = \frac{1}{N_{pos}N_{neg}}\sum_{i:y_i=1}\sum_{j:y_j=0}\mathbf{1}\left(s_i^{(t)} > s_j^{(t)}\right) \tag{13}$$

where N_{pos} and N_{neg} denote the numbers of anomalous and normal samples respectively, and $s_i^{(t)}$ the predicted anomaly score.

For regression, the coefficient of determination is reported as

$$R^2 = 1 - \frac{\sum_{i\in T}(\hat{y}_i - y_i)^2}{\sum_{i\in T}(y_i - \bar{y}_T)^2}, \quad \bar{y}_T = \frac{1}{N_T}\sum_{i\in T} y_i, \tag{14}$$

and root mean squared error is provided for completeness. The prompt templates used for inference can be serialized as JSON artifacts and reloaded verbatim to reproduce predictions under identical decoding settings, which underscores the distinction between prompt conditioned inference and parameter training.

This protocol differs fundamentally from machine learning and deep learning pipelines. In classical pipelines, data are consumed to estimate parameters through iterative optimization on a training split and validation is used for early stopping and hyperparameter selection. In the present setting, the foundation model retains its original weights and adapts through context alone. The few exemplars and task specifications embedded in the prompt act as conditioning evidence that steers domain specific reasoning at inference time. This leads to a distinct usage of data, since examples serve as contextual guidance rather than as gradient bearing samples. It also leads to a distinct cost profile, since compute is allocated to token processing rather than to parameter updates. Failure modes likewise differ. Parameterized models are sensitive to overfitting and underfitting, whereas prompt conditioned inference is sensitive to context length, exemplar selection, formatting, and decoding choices. For comparability, all approaches are evaluated on the same held out test partitions. Machine learning and deep learning baselines employ early stopping and hyperparameter search on the validation split, while the large language model performs inference without any parameter modification, and performance reflects only the information and constraints supplied by the prompt.

3. Results and discussion

3.1 Experiment settings

Experiments were conducted on a workstation equipped with an NVIDIA GeForce RTX 4070 Ti Super (16 GB). All large language model calls were issued via API. During entity extraction, qwen-turbo (2025-07-28) and qwen-plus (2025-07-15) were configured with temperature 0.1 to promote precise identification of research objects and task types.

For LLM based inference on spectra, temperature was fixed at 0.5 unless otherwise stated. Datasets followed a 6:2:2 split for training, validation, and testing. After initial benchmarking under this configuration, a backbone model was selected and applied across all datasets and tasks, with comparisons to standard machine learning and deep learning baselines. Ablation studies varied temperature and the number of training samples to quantify their effects on performance.

Preliminary trials indicated that regression labels often contain long decimal representations. When all samples are included, prompt length increases and can approach the context window limit, which may induce numeric hallucinations. To control context usage, the total sample count for regression experiments was capped at 100.

To ensure fair comparison, the 6:2:2 split was used consistently. The validation partition served as the early stopping criterion for deep learning baselines. Classical machine learning methods employed grid search for hyperparameter selection. The same training and test partitions were shared across LLM inference, machine learning baselines, and deep learning baselines so that performance differences reflect inference procedures rather than data selection. Each experiment was repeated ten times per material category, and results are reported as mean ± standard deviation.

3.2 Experiments of entity extraction and knowledge retrieval

3.2.1 Entity extraction accuracy

The performance of the entity extraction and subsequent knowledge retrieval modules was evaluated in order to assess the system's ability to identify both the research object and the analysis task from queries, and to retrieve the appropriate structured entries from the spectral literature corpus. For the entity extraction evaluation, a set of one hundred questions was manually

其中 pos 和 neg 分别表示异常样本和正常样本的数量， $s_i^{(t)}$ 为预测的异常分数。

对于回归任务，报告的决定系数为

$$R^2 = 1 - \frac{\sum_{i\in T}(\hat{y}_i - y_i)^2}{\sum_{i\in T}(y_i - \bar{y}_T)^2}, \quad \bar{y}_T = \frac{1}{N_T}\sum_{i\in T} y_i, \tag{14}$$

并提供均方根误差以确保完整性。用于推理的提示模板可序列化为 JSON 工件，并原样重新加载，以便在相同的解码设置下复现预测结果，这凸显了提示条件推理与参数训练之间的区别。

该协议与机器学习和深度学习流程存在根本差异。在传统流程中，数据被用于通过在训练集上进行迭代优化来估计参数，而验证集则用于早停和超参数选择。在当前设定下，基础模型保留其原始权重，仅通过上下文进行适配。嵌入提示中的少量示例和任务规范充当条件证据，引导推理时的领域特定推理。这导致了数据的独特用法，因为示例充当的是上下文指导，而非携带梯度的样本。这也导致了独特的成本结构，因为计算资源被分配给令牌处理，而非参数更新。失败模式 likewise 不同。参数化模型对过拟合和欠拟合敏感，而提示条件推理则对上下文长度、示例选择、格式化和解码选择敏感。为确保可比性，所有方法均在相同的保留测试分区上进行评估。机器学习和深度学习基线在验证集上采用早停和超参数搜索，而大语言模型则在不进行任何参数修改的情况下执行推理，其性能仅反映提示所提供的信息和约束。

3. 结果与讨论

3.1 实验设置

实验在配备 NVIDIA GeForce RTX 4070 Ti Super (16 GB) 的工作站上进行。所有大型语言模型调用均通过 API 发出。在实体提取过程中，qwen-turbo (2025-07-28) 和 qwen-plus (2025-07-15) 的温度参数设置为 0.1，以促进对研究对象和任务类型的精确识别。

对于基于大型语言模型的光谱推理，除非另有说明，温度参数固定为 0.5。数据集按 6:2:2 的比例划分为训练集、验证集和测试集。在此配置下进行初步基准测试后，选定一个骨干模型并将其应用于所有数据集和任务，同时与标准的机器学习和深度学习基线进行比较。消融研究通过改变温度参数和训练样本数量，以量化它们对性能的影响。

初步试验表明，回归标签通常包含较长的小数表示。当包含所有样本时，提示长度会增加并可能接近上下文窗口限制，从而可能引发数值幻觉。为了控制上下文使用量，回归实验的总样本数量上限设定为 100。

为确保公平比较，始终采用 6:2:2 的数据划分比例。验证集用作深度学习基线模型的早停准则。经典机器学习方法则采用网格搜索进行超参数选择。LLM 推理、机器学习基线与深度学习基线共享相同的训练集和测试集划分，从而确保性能差异反映的是推理过程的不同，而非数据选择的偏差。每项实验针对每种材料类别重复十次，结果以均值 ± 标准差的形式报告。

3.2 实体抽取与知识检索实验

3.2.1 实体抽取准确率

为了评估系统从查询中识别研究对象和分析任务，并从光谱文献语料库中检索相应结构化条目的能力，对实体提取及后续知识检索模块的性能进行了评估。针对实体提取的评估，人工构建了一百个问题，

constructed, covering three task types (classification, anomaly detection, regression) across five material categories. Each question was annotated with the ground truth research object label and task type label prior to testing. The prompts for entity extraction were then submitted to the qwen-turbo and qwen-plus models via API, with the research object entity evaluated using a fuzzy matching criterion (FuzzyWuzzy) and the task type entity assessed by exact match accuracy[26].

Table 2 Evaluation results of entity extraction

Models	Acc(Research object)	Acc(Task type)
qwen-turbo	82	89
qwen-plus	86	100

The results, summarized in Table 2, indicate that qwen-plus outperformed qwen-turbo on both metrics. Specifically, qwen-plus achieved an accuracy of 86% for extracting research object, compared to 82% for qwen-turbo, and correctly identified the task type in all cases (100% accuracy), whereas qwen-turbo attained 89% on this metric. These findings demonstrated that the enhanced language understanding capabilities of qwen-plus yield more precise entity recognition, particularly for distinguishing among closely related spectral materials.

Following successful entity extraction, the identified labels of research objects were used to query the structured knowledge base via retrieval algorithms. Manual inspection of a subset of retrieval outputs confirmed that over 90% of the top ranked documents contained metadata entries corresponding to the intended material, thereby validating the robustness of the retrieval pipeline. This end-to-end evaluation of entity extraction and knowledge retrieval establishes a reliable foundation for the downstream multi-task reasoning agents.

3.2.2 **Retrieval effectiveness**

In this study, three established lexical retrieval models were employed to identify relevant publications from the structured corpus: the Bag of Words (BoW) model, BM25, and TF-IDF cosine similarity. The BoW model represents each document as an unordered multiset of word frequencies, disregarding syntactic structure and term dependencies[27]. While straightforward to implement, its inability to capture contextual nuances often yields suboptimal precision in specialized domains. BM25, or Best Matching 25, builds upon the probabilistic retrieval framework by incorporating term frequency saturation, document length normalization, and inverse document frequency weighting; it has become a de facto standard in information retrieval for its balanced treatment of term importance and document verbosity[28]. The TF-IDF cosine similarity approach combines classical term frequency inverse document frequency weighting with a vector space model, measuring the cosine of the angle between query and document vectors to quantify their alignment; this method effectively emphasizes domain specific vocabulary while mitigating the influence of overly common terms.

Retrieval performance was evaluated on a curated test corpus of 200 documents sampled from the full SDAAP repository. One hundred queries were derived from the standardized research object labels produced in Section 3.2.1. For each query, the three highest ranked documents were retrieved by each model and compared against a set of preassigned relevance labels. Retrieval precision was calculated as the fraction of correctly retrieved documents among the top three for each query, and the mean precision across all queries was reported.

Table 3 Evaluation results of knowledge retrieval

Methods	Accuracy(%)
Bag of Words model	56.00
BM25	80.67
TF-IDF cosine similarity retrieval method	81.33

Table 3 summarizes the quantitative outcomes of these retrieval experiments. As shown, the Bag of Words model approach achieved a mean precision of 56.00%, reflecting the limitations of raw term counts in this specialized domain. The TF IDF model produced a substantially higher mean precision of 81.33%, demonstrating the benefit of inverse document frequency weighting in discriminating domain specific terms. BM25 attained a comparable mean precision of 80.67%, indicating that its term

涵盖五种材料类别下的三种任务类型（分类、异常检测、回归）。在测试之前，每个问题均标注了真实的研究对象标签和任务类型标签。随后，通过 API 将用于实体提取的提示提交给 qwen-turbo 和 qwen-plus 模型，其中研究对象实体采用模糊匹配标准（FuzzyWuzzy）进行评估，任务类型实体则通过精确匹配准确率进行评估[26]。

表 2 实体抽取的评估结果

模型	准确率（研究对象）	准确率（任务类型）
qwen-turbo	82	89
qwen-plus	86	100

表 2 汇总的结果表明，qwen-plus 在两项指标上均优于 qwen-turbo。具体而言，qwen-plus 在提取研究对象方面的准确率为 86%，而 qwen-turbo 为 82%；在任务类型识别方面，qwen-plus 实现了全覆盖（准确率 100%），而 qwen-turbo 在该指标上的准确率为 89%。这些发现证明，qwen-plus 增强的语言理解能力能够带来更精确的实体识别，尤其在区分密切相关的光谱材料方面表现突出。

在成功完成实体提取后，利用所识别的研究对象标签，通过检索算法查询结构化知识库。对部分检索结果的人工核查证实，超过 90% 的排名靠前的文档包含与目标材料相对应的元数据条目，从而验证了检索流程的鲁棒性。此次对实体提取与知识检索的端到端评估，为下游多任务推理智能体奠定了可靠基础。

3.2.2 **检索效果**

本研究采用了三种成熟的词汇检索模型，从结构化语料库中识别相关文献：词袋模型（BoW）、BM25 以及 TF-IDF 余弦相似度。词袋模型将每篇文档表示为词频的无序多重集合，忽略句法结构和词语依赖关系[27]。尽管该模型易于实现，但其无法捕捉上下文细微差别，往往导致在专业领域中的精度不佳。BM25（Best Matching 25）基于概率检索框架构建，引入了词频饱和机制、文档长度归一化以及逆文档频率加权；由于其对词语重要性和文档冗长度进行了平衡处理，已成为信息检索领域的事实标准[28]。TF-IDF 余弦相似度方法将经典的词频 - 逆文档频率加权与向量空间模型相结合，通过计算查询向量与文档向量之间夹角的余弦值来量化二者的对齐程度；该方法能有效突出领域特定词汇，同时降低过于常见词语的影响。

检索性能在从完整 SDAAP 存储库中采样的 200 份文档组成的精选测试语料库上进行了评估。一百个查询源自第 3.2.1 节生成的标准化研究对象标签。对于每个查询，各模型检索出排名最高的三份文档，并与一组预先分配的相关性标签进行对比。检索精确度计算为每个查询前三位中正确检索文档的比例，并报告了所有查询的平均精确度。

表 3 知识检索评估结果

方法	准确率 (%)
词袋模型	56.00
BM25	80.67
TF-IDF 余弦相似度检索方法	81.33

表 3 总结了这些检索实验的定量结果。如图所示，词袋模型方法的平均精确率为 56.00%，反映了原始词频计数在这一专业领域中的局限性。TF-IDF 模型产生了显著更高的平均精确率（81.33%），证明了逆文档频率加权在区分领域特定术语方面的优势。BM25 达到了与之相当的平均精确率（80.67%），表明其词项

frequency saturation and document length normalization confer performance similar to TF IDF under these pilot conditions. Together, these results confirm that weighted retrieval schemes markedly outperform the unweighted Bag of Words approach in identifying relevant spectral analysis literature within a noisy corpus.

3.3 Spectral Preprocessing and Feature Extraction Outcomes

The preprocessing and feature extraction pipeline for each material follows literature guided routines summarized in Table 4. For the public Milk dataset, a SG smoothing step is applied before principal component analysis, while Chinese medicinal herbs use SNV followed by a FD filter prior to PCA, in order to enhance weak band structure and reduce scattering effects. Citri Reticulatae Pericarpium employs SNV with subsequent PCA for dimensionality reduction. The private wastewater dataset is baseline corrected, implemented as asymmetric least squares in our workflow, and then encoded as Pearson correlation features against reference chemical oxygen demand profiles. Tecator and Corn, both public datasets, are standardized with SNV and then represented by partial least squares latent variables for downstream modeling. These choices are exactly those recorded in the updated retrieval Table 4 for materials, preprocessing, and feature extraction methods.

Table 4 Retrieval results for experimental material related preprocessing methods and feature extraction methods

Materials	Preprocess method	Feature extract method
Milk	SG	PCA
Chinese medicinal herbs	SNV+FD	PCA
CRP	SNV	PCA
Wastewater	BC	Pearson correlation features
Tecator	SNV	PLS
Corn	SNV	PLS

Figure 5 illustrates the effect of these choices by juxtaposing raw and processed spectra with their corresponding feature maps. On Milk, smoothing suppresses high frequency noise and stabilizes the variance structure for PCA. In Chinese medicinal herbs, the combination of SNV and FD sharpening reveals subtle curvature associated with overlapping constituents. For Citri Reticulatae Pericarpium, SNV aligns baselines and improves comparability across samples, enabling principal components to capture the dominant chemical variance. In wastewater, removal of baseline drift by asymmetric least squares prepares the signal for correlation based compression relative to reference profiles. For Tecator and Corn, SNV precedes partial least squares, which yields a compact set of latent factors that summarize the predictive spectral subspace. Together these outcomes confirm that the metadata driven selection procedure produces well conditioned and low dimensional representations that are consistent with the updated table and optimized for subsequent inference.

频率饱和与文档长度归一化在这些试点条件下赋予了与 TF-IDF 相似的性能。综上所述，这些结果证实，加权检索方案在从噪声语料中识别相关光谱分析文献方面，显著优于未加权的词袋方法。

3.3 光谱预处理与特征提取结果

每种材料的预处理和特征提取流程均遵循文献指导的常规方法，总结于表 4。对于公开的牛奶数据集，在主成分分析之前应用了 SG 平滑步骤；而中药材则先使用 SNV，再经过 FD 滤波后进行 PCA，以增强弱波段结构并减少散射效应。陈皮采用 SNV 并结合随后的 PCA 进行降维。私有废水数据集首先进行基线校正（在我们的工作流中实现为不对称最小二乘法），然后编码为相对于参考化学需氧量剖面的皮尔逊相关特征。Tecator 和 Corn 这两个公开数据集均通过 SNV 进行标准化，随后由偏最小二乘潜变量表示，用于下游建模。这些选择与更新后的检索表 4 中记录的材料、预处理及特征提取方法完全一致。

表 4 实验材料相关的预处理方法与特征提取方法的检索结果

材料	预处理方法	特征提取方法
Milk	SG	PCA
中药材	SNV+FD	PCA
CRP	SNV	PCA
废水	BC	皮尔逊相关特征
Tecator	SNV	PLS
Corn	SNV	PLS

图 5 通过并置原始光谱与处理后的光谱及其对应的特征图，展示了这些选择的效果。在牛奶样本中，平滑处理抑制了高频噪声并稳定了方差结构，从而利于主成分分析（PCA）。在中药材样本中，标准正态变量变换（SNV）与一阶导数（FD）锐化的结合揭示了与重叠成分相关的细微曲率变化。对于陈皮样本，SNV 对齐了基线并提高了样本间的可比性，使得主成分能够捕捉主要的化学变异。在废水样本中，采用非对称最小二乘法去除基线漂移，为基于与参考谱图相关性的信号压缩做好了准备。对于 Tecator 和 Corn 数据集，SNV 作为偏最小二乘法的前置步骤，生成了一组紧凑的潜变量因子，用以总结具有预测能力的光谱子空间。综上所述，这些结果证实，基于元数据的选择流程能够产生条件良好且低维的表示形式，这与更新后的表格一致，并针对后续推断进行了优化。

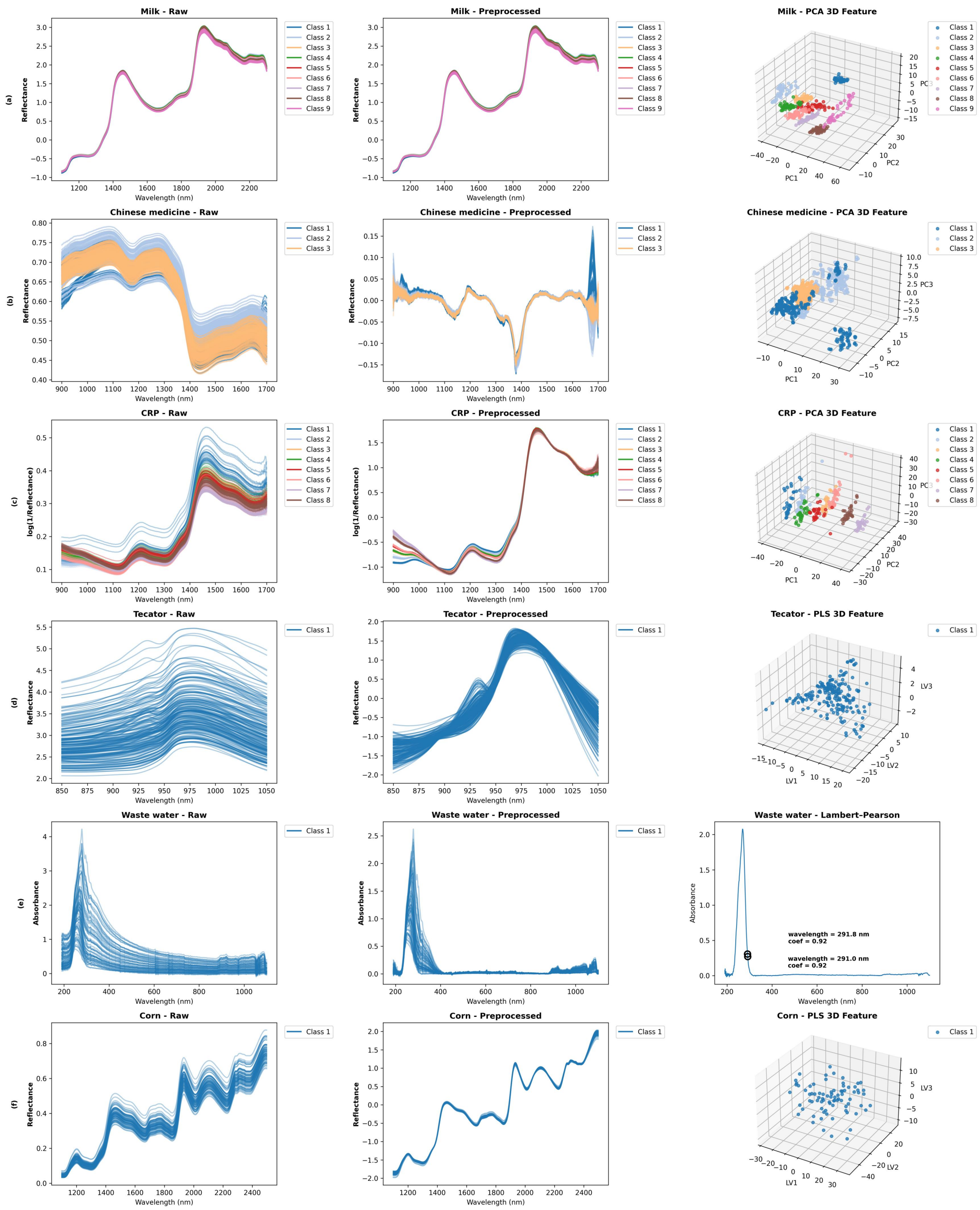


Figure 5. Raw spectra, preprocessed spectra, and resulting feature maps for six material categories: (a) Milk dataset, (b) Chinese medicinal herbs, (c) CRP (d) Tecator dataset, (e) Wastewater, (f) Corn dataset,

3.4 Experiments of multi-task reasoning

3.4.1 Model selection

This subsection identifies the default model for the proposed pipeline by jointly considering two complementary dimensions. Within this evaluation, four contemporary conversational models are examined. The Qwen series developed by Alibaba and accessed through the Alibaba Cloud API includes qwen-plus and qwen-turbo. Both adopt a Transformer backbone and are pretrained on large scale text and multimodal corpora to generate coherent, semantically rich outputs across domains. Qwen-plus, evaluated as the 2025-07-28 release, offers a balanced tradeoff among reasoning performance, cost, and latency, and is suitable for tasks of moderate complexity. Qwen-turbo, evaluated as the 2025-07-15 release, prioritizes inference speed and minimal cost, with correspondingly reduced reasoning depth for simpler workloads[29]. DeepSeek-v3, released on December 26,

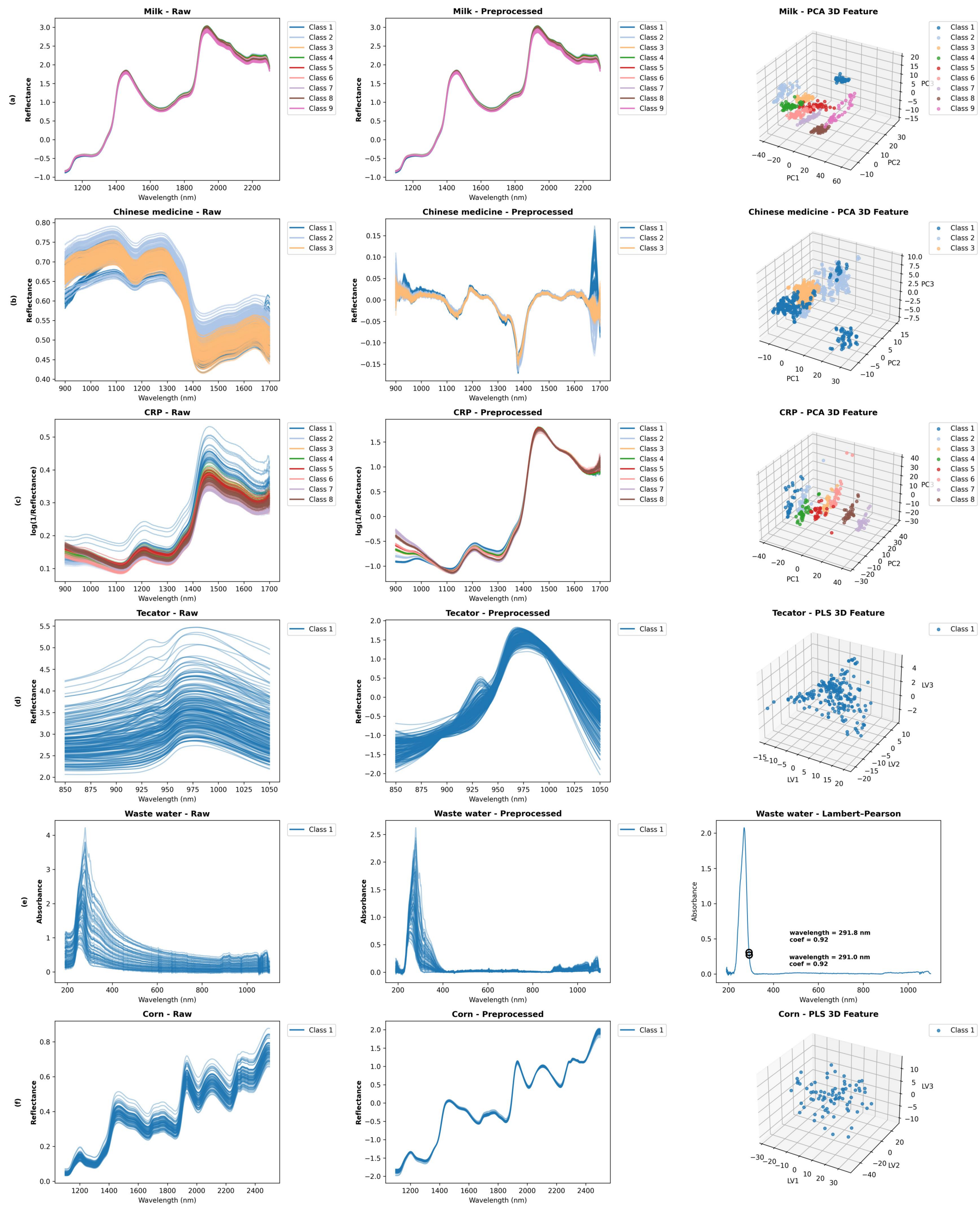


图 5. 六种材料类别的原始光谱、预处理光谱及生成的特征图：(a) 牛奶数据集，(b) 中药材，(c) CRP，(d) Tecator 数据集，(e) 废水，(f) 玉米数据集，

3.4 多任务推理实验

3.4.1 模型选择

本小节通过综合考量两个互补维度，确定了所提出流程的默认模型。在此次评估中，考察了四种当代对话模型。由阿里巴巴开发并通过阿里云 API 提供的通义千问（Qwen）系列包括 qwen-plus 和 qwen-turbo。两者均采用 Transformer 架构，并在大规模文本及多模态语料库上进行预训练，以跨领域生成连贯且语义丰富的输出。qwen-plus（评估版本为 2025-07-28 发布）在推理性能、成本和延迟之间提供了均衡的权衡，适用于中等复杂度的任务。qwen-turbo（评估版本为 2025-07-15 发布）则优先保障推理速度和最低成本，相应地降低了针对简单工作负载的推理深度[29]。DeepSeek-v3 发布于 12 月 26 日，

2024 by DeepSeek, is a mixture of experts model with 671 billion parameters pretrained on 14.8 trillion tokens, targeting high capacity reasoning at scale[30]. ChatGPT-5, accessed as chatgpt-5-chat-latest from OpenAI, is a high capacity conversational model finetuned for instruction following and multi turn dialogue, and serves here as a strong general purpose baseline[31].

The first dimension comprises practical constraints that directly affect inference, including the effective context window exposed by each API, the cost per million tokens under provider pricing, and the average completion length required by each task. In this setting, classification yields short label strings, anomaly detection returns binary outputs, and regression produces substantially longer numeric sequences. To characterize token usage, the number of output tokens generated during inference was recorded. Results are presented in Table 5.

Table 5 Comparison of LLMs under generation constraints

LLMs	Classification Output tokens	Regression Output tokens	Anomaly detection Output tokens	Context window limitations	Cost per million tokens
qwen-plus	135	112	25	1000K	0–128K：¥2，128–256K：¥20，256K–1M：¥48
qwen-turbo	failed	112	25	1000K	¥6
deepseek-v3	144	81	25	64K	¥0.5
chatgpt-5-chat	146	81	26	400K	$10

The second dimension comprises empirical performance across tasks. For each model, outcomes were measured on shared dataset splits with matched prompts and temperature schedules, and the task metrics described above were summarized as means and dispersion across repeated runs. The benchmark includes qwen-plus, qwen-turbo, deepseek-v3, and chatgpt-5-chat-latest, evaluated on the same task suite, with Milk used for classification, Chinese medicinal herbs for anomaly detection, and Corn for regression. Experimental results are reported in Table 6. Instances marked as failed in Tables 5 and 6 are primarily attributable to the large sample size in the classification task, which exceeded the operational capacity of qwen-turbo in this setting and prevented the generation of valid classification outputs.

Table 6 Experiment results of different LLMs

LLMs	Classification (Milk)	Regression (Corn)		Anomaly detection (Chinese medicinal herbs)	
	Acc (%)	R2	RMSE	Precision	AUC
qwen-plus	69.72±3.7	0.8117±0.0351	0.2258±0.0214	1±0	1±0
qwen-turbo	failed	0.7320±0.0529	0.2693±0.0261	1±0	0.75±0
deepseek-v3	50.83±5.69	0.6731±0.0588	0.2976±0.0274	1±0	1±0
chatgpt-5-chat	**96.10±1.52**	**0.9252±0.0065**	**0.1427±0.0063**	**1±0**	**1±0**

Across all three tasks ChatGPT-5 provides the most robust and accurate performance. Although its unit price per generated token is higher, the predictions studied here require very few tokens because outputs are short labels, compact numbers, or binary decisions. The effective cost per example is therefore modest while accuracy remains consistently high. Consequently, ChatGPT-5 is selected as the default model in the pipeline, and all subsequent experiments and ablations use ChatGPT-5 for inference.

3.4.2 Results of Classification

This task aims to evaluate the capability of different models to accurately classify spectroscopic samples from three material categories: milk, Chinese medicinal materials, and Citri Reticulatae Pericarpium (CRP). Each sample's input comprises raw spectral data subjected to category specific optimal preprocessing techniques, followed by feature extraction using PCA with five retained components. These low dimensional spectral features were used as inputs for both LLM based models and classical

2024 年由 DeepSeek 推出的模型是一个拥有 6710 亿参数的混合专家模型，在 14.8 万亿 token 上进行了预训练，旨在实现大规模的高容量推理[30]。ChatGPT-5（通过 OpenAI 以 chatgpt-5-chat-latest 形式访问）是一个高容量的对话模型，经过微调以遵循指令并支持多轮对话，在此作为强大的通用基线[31]。

第一个维度包含直接影响推理的实际约束条件，包括每个 API 提供的有效上下文窗口、提供商定价下每百万 token 的成本，以及每项任务所需的平均完成长度。在此设定下，分类任务生成简短的标签字符串，异常检测返回二元输出，而回归任务则产生明显更长的数值序列。为了表征 token 使用情况，我们记录了推理过程中生成的输出 token 数量。结果见表 5。

表 5 生成约束下的 LLM 比较

LLMs	分类 输出令牌	回归 输出令牌	异常检测 输出令牌	上下文窗口限制	每百万令牌成本
qwen-plus	135	112	25	1000K	0–128K：¥2，128–256K：¥20，256K–1M：¥48
qwen-turbo	失败	112	25	1000K	¥6
deepseek-v3	144	81	25	64K	¥0.5
chatgpt-5-chat	146	81	26	400K	$10

第二个维度涵盖各任务上的实证表现。对于每个模型，我们在共享的数据集划分上测量结果，使用匹配的提示词和温度调度，并将上述任务指标汇总为多次运行的均值和离散度。该基准测试包括 qwen-plus、qwen-turbo、deepseek-v3 和 chatgpt-5-chat-latest，均在相同的任务套件上进行评估：其中 Milk 用于分类任务，中药材用于异常检测任务，Corn 用于回归任务。实验结果见表 6。表 5 和表 6 中标记为失败的实例，主要归因于分类任务中的样本量过大，超出了 qwen-turbo 在此设置下的处理能力，导致无法生成有效的分类输出。

表 6 不同大语言模型的实验结果

LLMs	分类（Milk）	回归（Corn）		异常检测（中药材）	
	准确率 (%)	R2	RMSE	精度	AUC
qwen-plus	69.72±3.7	0.8117±0.0351	0.2258±0.0214	1±0	1±0
qwen-turbo	失败	0.7320±0.0529	0.2693±0.0261	1±0	0.75±0
deepseek-v3	50.83±5.69	0.6731±0.0588	0.2976±0.0274	1±0	1±0
chatgpt-5-chat	**96.10±1.52**	**0.9252±0.0065**	**0.1427±0.0063**	**1±0**	**1±0**

在所有三项任务中，ChatGPT-5 提供了最稳健且准确的性能。尽管其每生成 token 的单价较高，但本研究中的预测仅需极少的 token，因为输出为简短标签、紧凑数值或二元决策。因此，每个示例的有效成本较低，同时准确率始终保持高位。鉴于此，ChatGPT-5 被选为流程中的默认模型，所有后续实验与消融研究均使用 ChatGPT-5 进行推理。

3.4.2 分类结果

本任务旨在评估不同模型准确分类来自三类材料（牛奶、中药材及陈皮）的光谱样本的能力。每个样本的输入包含经过类别特定最优预处理技术处理的原始光谱数据，随后利用主成分分析（PCA）提取特征并保留五个分量。这些低维光谱特征被用作基于大语言模型（LLM）的模型和经典

machine learning classifiers.

As baselines, three classical classifiers were tested: support vector machine (SVM)[32], k-nearest neighbors (KNN)[33], and random forest (RF) which are all trained on the same features extracted by PCA[34]. We further evaluated a one dimensional convolutional neural network (CNN)[35] and a simple Transformer encoder[36], both trained on the raw, preprocessed spectral vectors.

Table 7 Experiment results of classification task

Methods	Milk Acc.(%)	Chinese medicinal herbs Acc.(%)	CRP Acc.(%)
LLM	96.10±1.52	100±0	99.58±0.93
SVM	100±0	100±0	95.83±0
KNN	93.06±0	100±0	89.58±0
RF	97.22±0	100±0	95.83±0
CNN	95.83±0.98	90±14	91.25±1.75
Transformer	95.11±3.72	85±15	97.50±3.42

As shown in Table 7, across the three datasets, distinct performance patterns emerge among classical machine learning models, deep architectures, and the large language model (LLM). On the Milk dataset, SVM achieves perfect accuracy, while the LLM and random forest also perform competitively. CNN and Transformer yield slightly lower but still strong results, whereas KNN lags behind, reflecting its sensitivity to data distribution complexity.

For Chinese medicinal herbs, the task appears relatively simple, as most methods (LLM, SVM, KNN, RF) achieve perfect classification. By contrast, CNN and Transformer exhibit noticeable variability, suggesting that high capacity neural architectures are prone to instability under limited data conditions.

On the CRP dataset, the LLM attains the highest accuracy, surpassing both classical and deep learning baselines. Transformer also performs strongly, whereas SVM and RF deliver moderate results, and KNN again underperforms .

Taken together, these results indicate that the LLM consistently provides robust and accurate predictions across tasks, excelling particularly in more complex scenarios such as CRP. While SVM and RF remain competitive in simpler settings, their advantages diminish when nonlinearities increase. CNN and Transformer demonstrate potential but exhibit higher variance, reflecting a trade off between capacity and stability in small to medium scale data regimes.

3.4.3 **Results of Regression**

In the regression study we evaluate linear regression, support vector regression[37], and partial least squares regression on reduced features[38], together with random forest and two end to end neural baselines, a one dimensional convolutional network and a compact Transformer, trained on preprocessed spectra. The large language model conducts direct single shot inference from exemplar augmented prompts without any parameter updates. For Tecator and Corn, which are public datasets, features are obtained through partial least squares with four components, while the wastewater dataset, which is private, uses a small set of physically interpretable features derived from absorbance laws and correlation structure. Table 8 summarizes the results in terms of coefficient of determination and root mean squared error.

Table 8 Experiment results of regression task

Type	Methods	R2	RMSE
Tecator	LLM	0.9592±0.0033	2.9221±0.1180
	LR	0.9696±0	2.5234±0
	SVR	0.9876±0	1.6100±0
	PLSR	0.9696±0	2.5234±0

机器学习分类器的输入。

作为基线，测试了三种经典分类器：支持向量机（SVM）[32]、K 近邻（KNN）[33],和随机森林（RF），它们均在通过 PCA 提取的相同特征上进行训练[34]。我们还评估了一维卷积神经网络（CNN）[35] 和简单的 Transformer 编码器[36],，两者均在原始的预处理光谱向量上进行训练。

表 7 分类任务的实验结果

方法	牛奶准确率 (%)	中草药准确率 (%)	CRP 准确率 (%)
LLM	96.10±1.52	100±0	99.58±0.93
SVM	100±0	100±0	95.83±0
KNN	93.06±0	100±0	89.58±0
RF	97.22±0	100±0	95.83±0
CNN	95.83±0.98	90±14	91.25±1.75
变压器	95.11±3.72	85±15	97.50±3.42

如表 7 所示，在三个数据集中，经典机器学习模型、深度架构和大语言模型（LLM）呈现出不同的性能模式。在 Milk 数据集上，SVM 实现了完美的准确率，而 LLM 和随机森林也表现优异。CNN 和 Transformer 的结果略低但仍强劲，而 KNN 则落后，反映了其对数据分布复杂性的敏感性。

对于中药材任务，其难度似乎相对较低，因为大多数方法（LLM、SVM、KNN、RF）均实现了完美分类。相比之下，CNN 和 Transformer 表现出明显的波动性，表明高容量神经架构在数据有限的条件下容易不稳定。

在 CRP 数据集上，LLM 取得了最高准确率，超越了经典学习和深度学习基线。Transformer 同样表现强劲，而 SVM 和 RF 交付了中等结果，KNN 再次表现不佳。

综上所述，这些结果表明，LLM 在各任务中始终提供稳健且准确的预测，尤其在 CRP 等更复杂的场景中表现卓越。虽然 SVM 和 RF 在较简单的设置中仍具竞争力，但随着非线性增加，其优势逐渐减弱。CNN 和 Transformer 展现了潜力，但表现出更高的方差，反映了在中小规模数据 regime 下容量与稳定性之间的权衡。

3.4.3 **回归结果**

在回归研究中，我们在降维特征[37], 上评估了线性回归、支持向量回归[38], 和偏最小二乘回归，并结合随机森林以及两种端到端神经基线模型（一维卷积网络和紧凑型 Transformer），这些模型均在预处理光谱上进行训练。大语言模型基于示例增强提示执行直接单次推理，无需任何参数更新。对于公共数据集 Tecator 和 Corn，特征通过包含四个成分的偏最小二乘法获得；而私有数据集 wastewater 则使用一组源自吸收定律和相关结构的小型物理可解释特征。表 8 总结了决定系数和均方根误差方面的结果。

表 8 回归任务的实验结果

Type	方法	R2	RMSE
Tecator	LLM	0.9592±0.0033	2.9221±0.1180
	LR	0.9696±0	2.5234±0
	SVR	0.9876±0	1.6100±0
	PLSR	0.9696±0	2.5234±0

	RF	0.9551±0	3.0674±0
	CNN	0.9796±0.0062	2.0486±0.3117
	Transformer	0.8909±0.0726	4.5784±1.5438
Corn	LLM	0.9252±0.0065	0.1427±0.0063
	LR	0.8826±0	0.1789±0
	SVR	0.9096±0	0.1570±0
	PLSR	0.8850±0	0.1771±0
	RF	0.6474±0	0.3101±0
	CNN	0.7938±0.0365	0.2364±0.0212
	Transformer	0.7262±0.0874	0.2708±0.0411
Wastewater	LLM	0.9185±0.0066	12.2362±0.5011
	LR	0.7409±0	21.8310±0
	SVR	0.8277±0	17.8029±0
	PLSR	0.7447±0	21.6732±0
	RF	0.8013±0	19.1181±0
	CNN	0.7535±0.0804	21.0832±3.3542
	Transformer	0.9021±0.0655	12.8519±4.3135

Across datasets the ranking of methods is consistent with the interplay between feature design, sample size, and nonlinearity. On Tecator the strongest performance is achieved by a kernel based method, indicating that the residual mapping after partial least squares reduction remains smooth yet meaningfully nonlinear in a way that a radial basis function can capture with low variance. The convolutional baseline is also competitive, which suggests that local spectral patterns are stable enough for localized filters to generalize well. Linear models on the reduced space remain adequate but not optimal, and the Transformer shows the widest variability, a sign of sensitivity to initialization and limited data when model capacity is high. The large language model is close to but not at the top on this dataset, which is consistent with a setting where kernel smoothing is well matched to the structure of the remaining nonlinearity.

On Corn the large language model attains the best overall balance of higher coefficient of determination and lower error, outperforming both linear baselines and the kernel method on the same reduced features. This pattern implies that the mapping in the reduced space departs from linearity in a way that is better captured by the prompt conditioned function learned from exemplars. End to end deep models underperform in this regime, which points to a data requirement that is not fully met when training directly on full spectra. The stability of the large language model across repeats further indicates that exemplar conditioning on low dimensional, domain aware features can deliver both accuracy and robustness without parameter tuning.

On the private wastewater dataset the large language model again leads, followed by the Transformer, while classical baselines are clearly behind. Here the features encode physical constraints and pairwise relations, and the advantage of the large language model suggests that its prompt conditioned mapping can approximate the moderate nonlinearity expressed by these interpretable features more effectively than linear predictors and with less variance than a high capacity Transformer. Random forest and the convolutional baseline do not benefit as much from these compact features, which is consistent with their preference for richer representation learning directly from high dimensional inputs.

Taken together these observations offer a simple practical recipe. When domain informed reduction yields compact and informative features, direct large language model inference is a strong default that combines accuracy with stability and very low engineering overhead. If validation indicates a very smooth residual structure in the reduced space, a kernel method can still be preferable. End to end neural models remain valuable when data are abundant and the goal is to exploit fine grained structure in raw spectra, but in the small to medium sample regime they are more sensitive to initialization and regularization. The contrast between the two public datasets and the private wastewater dataset also highlights the role of feature construction. Public

	RF	0.9551±0	3.0674±0
	CNN	0.9796±0.0062	2.0486±0.3117
	变压器	0.8909±0.0726	4.5784±1.5438
Corn	LLM	0.9252±0.0065	0.1427±0.0063
	LR	0.8826±0	0.1789±0
	SVR	0.9096±0	0.1570±0
	PLSR	0.8850±0	0.1771±0
	RF	0.6474±0	0.3101±0
	CNN	0.7938±0.0365	0.2364±0.0212
	变压器	0.7262±0.0874	0.2708±0.0411
废水	LLM	0.9185±0.0066	12.2362±0.5011
	LR	0.7409±0	21.8310±0
	SVR	0.8277±0	17.8029±0
	PLSR	0.7447±0	21.6732±0
	RF	0.8013±0	19.1181±0
	CNN	0.7535±0.0804	21.0832±3.3542
	变压器	0.9021±0.0655	12.8519±4.3135

在各数据集中，方法的排名与特征设计、样本量及非线性之间的相互作用保持一致。在 Tecator 数据集上，表现最佳的是基于核的方法，这表明经偏最小二乘降维后的残差映射仍然平滑，但具有径向基函数能够以低方差捕捉的显著非线性。卷积基线模型也具备竞争力，说明局部光谱模式足够稳定，使局部滤波器能够良好泛化。在降维空间上的线性模型表现尚可但非最优，而 Transformer 表现出最大的变异性，这反映出在高模型容量下对初始化和有限数据的敏感性。大语言模型在该数据集上的表现接近但未达到顶尖水平，这与核平滑方法能很好地匹配剩余非线性结构的场景一致。

在 Corn 数据集上，大语言模型实现了更高的决定系数与更低的误差之间的最佳整体平衡，在相同的降维特征上优于线性基线和核方法。这一模式表明，降维空间中的映射偏离了线性关系，而这种非线性能够被基于示例学习到的提示条件函数更好地捕捉。端到端深度模型在此场景下表现不佳，这表明直接在完整光谱上进行训练时，数据需求尚未得到充分满足。大语言模型在多次重复实验中的稳定性进一步表明，针对低维、领域感知特征的示例 conditioning 能够在无需参数调优的情况下，同时提供准确性和鲁棒性。

在私有废水数据集上，大语言模型再次领先，Transformer 紧随其后，而经典基线明显落后。此处特征编码了物理约束和成对关系，大语言模型的优势表明，其基于提示的映射能够比线性预测器更有效地近似这些可解释特征所表达的适度非线性，且相比高容量的 Transformer 具有更小的方差。随机森林和卷积基线从这些紧凑特征中获益较少，这与它们倾向于直接从高维输入进行更丰富的表示学习是一致的。

综上所述，这些观察结果提供了一个简单实用的方案。当基于领域知识的降维能够产生紧凑且信息丰富的特征时，直接进行大语言模型推理是一个强有力的默认选择，它兼具准确性、稳定性以及极低的工程开销。如果验证表明降维空间中的残差结构非常平滑，核方法可能仍然是更优的选择。端到端神经模型在数据充足且目标是利用原始光谱中细粒度结构时依然具有价值，但在中小样本量情况下，它们对初始化和正则化更为敏感。两个公共数据集与私有污水数据集之间的对比也突显了特征构建的作用。公共

benchmarks with partial least squares features can favor methods that capture smooth nonlinearities with small variance, whereas physically motivated features extracted from domain principles can amplify the gains of exemplar conditioned large language model inference.

3.4.4 Results of Anomaly Detection

This task investigates anomaly detection across three spectroscopic material categories: milk, Chinese medicinal herbs, and Citri Reticulatae Pericarpium (CRP). The input for each sample comprises spectroscopic data that has been preprocessed using the optimal spectral transformation for its material class, followed by dimensionality reduction via PCA with five components. This results in a five dimensional spectral feature vector per sample, which serves as the input to all models under evaluation.

For each material, a subclass was designated as the reference class and considered "normal," comprising 50% of the data. The remaining 50% was divided into two types of anomalies: 33.33% were taken from different subclasses (inter class anomalies), and 16.67% were generated by applying small perturbations to the reference data (intra class anomalies). All normal samples were labeled as True, while both anomaly types were labeled False. The model's task is to perform binary classification for each spectrum, indicating whether it conforms to the learned "normal" distribution. Evaluation metrics included precision and the area under the receiver operating characteristic curve (AUC), which together reflect detection accuracy and robustness under class imbalance.

As baselines, three classical one class anomaly detection models were also considered: Isolation Forest (IF), which uses random partitioning to isolate anomalies[39] and one class support vector machine (OC-SVM), which attempts to separate the origin from the data in feature space using a hyperplane[40]. To evaluate lightweight deep learning alternatives, we also tested an autoencoder trained on the normal class and an LSTM autoencoder cascade[41][42], both using the same spectral features.

Table 9 Experiment results of anomaly detection task

Methods	Milk		Chinese medicinal herbs		CRP	
	Precision	AUC	Precision	AUC	Precision	AUC
LLM	1±0	0.950±0.046	1±0	1±0	1±0	0.75±0
IF	0±0	0.500±0	0±0	0.500±0	0±0	0.5±0
OC-SVM	1±0	0.167±0	0.625±0	0.2778±0	1±0	0.125±0
Autoencoder	1±0	1±0	1±0	0.628±0.061	1±0	0.975±0.056
LSTM+Autoencoder	1±0	0.994±0.012	1±0	0.628±0.046	1±0	0.75±0

The results, summarized in Table 9, reveal distinct performance patterns across datasets and methods. The LLM demonstrates consistently high precision across all three materials, with particularly strong performance on Chinese medicinal herbs where both precision and AUC reach perfect scores even when the number of training samples is limited. This robustness suggests that the separation between normal and anomalous samples in this dataset is well captured by the LLM representations, allowing reliable detection even under minimal supervision. For the milk dataset, the LLM also achieves perfect precision and a stable AUC close to 0.95, outperforming conventional approaches that fail to detect anomalies reliably. On the CRP dataset, while precision remains perfect, the AUC is lower, indicating that although the model can classify anomalies correctly, its ability to rank samples by anomaly likelihood is more limited in this case.

Among the baselines, IF and OC-SVM exhibit poor detection capability, often defaulting to random or biased predictions. In contrast, the autoencoder achieves strong results on milk and CRP, showing that reconstruction-based strategies can capture meaningful deviations in spectral structure. However, its performance drops markedly on Chinese medicinal herbs, highlighting the dataset-dependent sensitivity of such approaches. The LSTM – autoencoder cascade improves the stability of AUC on milk but shows no substantial advantage on the other datasets, suggesting that increased model complexity does not necessarily

基准测试中采用偏最小二乘特征往往有利于那些能以较小方差捕捉平滑非线性的方法，而基于领域原理提取的物理驱动特征则能进一步放大基于示例条件的大语言模型推理的优势。

3.4.4 异常检测结果

本研究针对牛奶、中药材及柑橘皮（CRP）三类光谱材料开展异常检测。每个样本的输入数据均先经其所属材料类别的最优光谱变换预处理，随后通过主成分分析（PCA）降维至五个分量。由此，每个样本生成一个五维光谱特征向量，作为所有待评估模型的输入。

针对每种材料，指定一个子类作为参考类并视为“正常”样本，占数据总量的 50%。其余 50% 的数据划分为两类异常：33.33% 源自不同子类（类间异常），16.67% 则通过对参考数据施加微小扰动生成（类内异常）。所有正常样本标记为 True，两类异常样本均标记为 False。模型任务是对每条光谱执行二分类，判断其是否符合所学习的“正常”分布。评估指标包括精确率以及受试者工作特征曲线下的面积（AUC），二者共同反映在类别不平衡条件下的检测准确率与鲁棒性。

作为基线，我们还考虑了三种经典的单类异常检测模型：隔离森林（IF），它利用随机划分来隔离异常点[39]；以及单类支持向量机（OC-SVM），它试图在特征空间中通过超平面将原点与数据分离开来[40]。为了评估轻量级的深度学习替代方案，我们还测试了在正常类别上训练的自编码器以及 LSTM 自编码器级联模型[41][42]，，两者均使用相同的频谱特征。

表 9 异常检测任务的实验结果

方法	Milk		中草药		CRP	
	精度	AUC	精度	AUC	精度	AUC
LLM	1±0	0.950±0.046	1±0	1±0	1±0	0.75±0
IF	0±0	0.500±0	0±0	0.500±0	0±0	0.5±0
OC-SVM	1±0	0.167±0	0.625±0	0.2778±0	1±0	0.125±0
自编码器	1±0	1±0	1±0	0.628±0.061	1±0	0.975±0.056
LSTM+自编码器	1±0	0.994±0.012	1±0	0.628±0.046	1±0	0.75±0

表 9 汇总的结果揭示了不同数据集和方法之间显著的性能差异。大语言模型在所有三种材料上均展现出持续的高精度，尤其在中药材数据集中表现尤为突出：即使训练样本数量有限，其精确率和 AUC 仍达到完美分数。这种鲁棒性表明，大语言模型的表征能够很好地捕捉该数据集中正常样本与异常样本之间的区分特征，从而在极少监督条件下实现可靠检测。对于牛奶数据集，大语言模型同样实现了完美的精确率，并保持约 0.95 的稳定 AUC，优于那些无法可靠检测异常的常规方法。而在 CRP 数据集中，尽管精确率保持完美，AUC 却较低，这表明虽然模型能够正确分类异常样本，但其按异常可能性对样本进行排序的能力在此情况下较为有限。

在各基线方法中，IF 和 OC-SVM 的检测能力较差，往往倾向于随机或有偏的预测。相比之下，自编码器在牛奶和 CRP 数据集上取得了优异结果，表明基于重构的策略能够捕捉光谱结构中有意义的偏差。然而，其在中药材数据集上的性能显著下降，凸显了此类方法对数据集的敏感性。LSTM 与自编码器的级联模型提升了牛奶数据集上 AUC 的稳定性，但在其他数据集上并未展现出明显优势，这表明模型复杂度的增加并不一定

translate into more effective anomaly detection in this domain.

Overall, these findings confirm the ability of the LLM framework to deliver robust and accurate anomaly detection across diverse spectroscopic materials. Its superiority over traditional machine learning methods and competitive performance relative to specialized deep architectures underscore the value of leveraging language-model-based reasoning for complex chemometric tasks.

3.5 Ablation study

This section examines two design factors of the proposed agent—decoding temperature and the size of the training set—across three representative tasks: Milk classification, Corn regression, and anomaly detection on Chinese medicinal herbs. All trends are discussed with respect to Table 10 and Table 11.

3.5.1 Temperature

Temperature modulates the entropy of the LLM's token sampling and therefore governs the balance between determinism and exploratory diversity during reasoning. Table 10 reports the experimental results obtained under different temperature settings, covering classification, regression, and anomaly detection tasks.

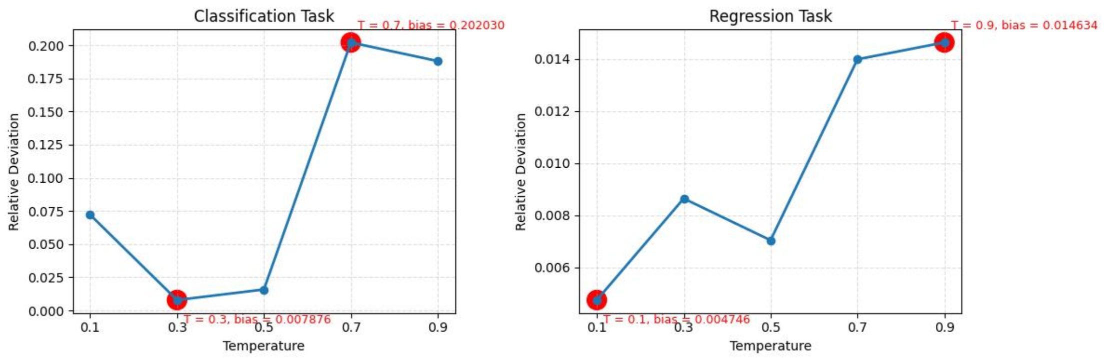
Table 10 Experiments results of different temperature

Temperature	Classification (Milk)	Regression (Corn)		Anomaly detection (Chinese medicinal herbs)	
	Acc (%)	R2	RMSE	Precision	AUC
0.1	86.10±6.21	0.9354±0.0044	0.1326±0.0046	1±0	1±0
0.3	96.66±0.76	0.9344±0.0081	0.1336±0.0083	1±0	1±0
0.5	96.10±1.52	0.9252±0.0065	0.1427±0.0063	1±0	1±0
0.7	89.16±18.01	0.9300±0.013	0.1377±0.0131	1±0	1±0
0.9	87.77±16.50	0.9192±0.0135	0.1480±0.0127	1±0	1±0

Table 10 illustrates how temperature affects model performance across classification, regression, and anomaly detection tasks. For Milk classification, accuracy improves as temperature increases from 0.1 to 0.3, reaching its best performance around 0.3-0.5, but then declines at higher temperatures. This pattern indicates that moderate randomness in sampling helps balance stability and exploration, while excessive determinism or stochasticity undermines classification robustness.

For Corn regression, the best results are observed at low temperatures. At 0.1, the model achieves the highest R2 and lowest RMSE, while performance degrades gradually as temperature increases, with R2 dropping and RMSE rising at 0.9. This trend highlights that regression tasks favor more deterministic outputs, as randomness introduces variance that weakens continuous value prediction.

In Chinese medicinal herbs anomaly detection, Precision and AUC remain consistently perfect across all temperatures, suggesting that the task is insensitive to sampling variations, likely due to clear separability in feature space.



能转化为该领域更有效的异常检测能力。

总体而言，这些发现证实了该大语言模型框架能够在多样化的光谱材料中实现稳健且准确的异常检测。其优于传统机器学习方法的表现，以及与专用深度架构相比具有竞争力的性能，凸显了利用基于语言模型的推理来处理复杂化学计量学任务的价值。

3.5 消融研究

本节考察了所提出代理的两个设计因素——解码温度和训练集规模——涉及三个代表性任务：牛奶分类、玉米回归以及中药材异常检测。所有趋势均结合表 10 和表 11 进行讨论。

3.5.1 温度

温度调节大语言模型令牌采样的熵，从而控制推理过程中确定性与探索性多样性之间的平衡。表 10 报告了在不同温度设置下获得的实验结果，涵盖分类、回归和异常检测任务。

表 10 不同温度下的实验结果

温度	分类（牛奶）	回归（玉米）		异常检测（中药材）	
	准确率 (%)	R2	RMSE	精度	AUC
0.1	86.10±6.21	0.9354±0.0044	0.1326±0.0046	1±0	1±0
0.3	96.66±0.76	0.9344±0.0081	0.1336±0.0083	1±0	1±0
0.5	96.10±1.52	0.9252±0.0065	0.1427±0.0063	1±0	1±0
0.7	89.16±18.01	0.9300±0.013	0.1377±0.0131	1±0	1±0
0.9	87.77±16.50	0.9192±0.0135	0.1480±0.0127	1±0	1±0

表 10 展示了温度如何影响模型在分类、回归和异常检测任务中的性能。对于牛奶分类任务，随着温度从 0.1 升高到 0.3，准确率随之提升，并在 0.3–0.5 区间达到最佳表现，随后在更高温度下开始下降。这一模式表明，适度的采样随机性有助于平衡稳定性与探索性，而过度的确定性或随机性则会削弱分类的鲁棒性。

对于玉米回归任务，最佳结果出现在较低温度下。在温度为 0.1 时，模型取得最高的 R^2 值和最低的 RMSE 值；随着温度升高，性能逐渐下降，至 0.9 时 R^2 降低而 RMSE 上升。这一趋势凸显了回归任务更倾向于确定性输出，因为随机性会引入方差，从而削弱连续值预测的准确性。

在中药材异常检测中，精确率（Precision）和 AUC 在所有温度下始终保持完美，表明该任务对采样变化不敏感，这可能是由于特征空间中具有清晰的可分性。

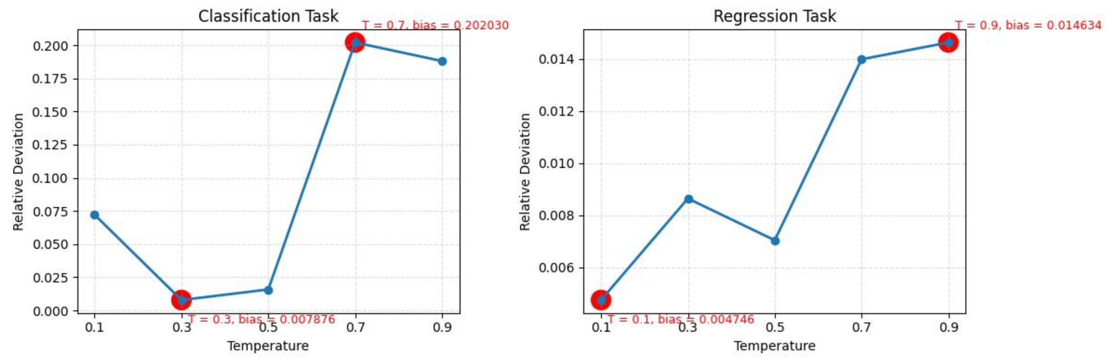


Figure 6. Temperature ablation analysis of relative average deviation (RAD) for classification and regression tasks.

To further validate the impact of temperature, we additionally plot the relative average deviation (RAD) curves in Figure 6. For classification, the RAD reaches its minimum at T = 0.3 and maximum at T = 0.7, showing that moderate temperatures stabilize the model's predictions while higher temperatures lead to unstable deviations. For regression, RAD is lowest at T = 0.1 and highest at T = 0.9, reinforcing that regression tasks require lower temperatures to minimize prediction variance. Together, the RAD analysis quantitatively confirms that classification benefits from moderate stochasticity, regression favors low temperatures, and anomaly detection remains stable regardless of temperature.

Taken together, these findings suggest that optimal temperature settings depend on task type: classification benefits from moderate value, regression favors lower values, and anomaly detection remains robust regardless of temperature.

3.5.2 Size of dataset

Varying the number of training samples reveals distinct data efficiency profiles for the three tasks, while keeping the test set size fixed during experiments. Table 11 presents the impact of varying training set sizes on model performance for classification, regression, and anomaly detection.

Table 11 Experiment results of different size of dataset

Task	Size of Training set	Metrics	
Classification (Milk)		Acc (%)	
	72	30.83±3.85	
	144	47.50±3.00	
	216	96.10±1.52	
Regression (Corn)		R2	RMSE
	16	0.8354±0.0236	0.2114±0.0152
	32	0.8472±0.0281	0.2035±0.0184
	48	0.9252±0.0065	0.1427±0.0063
Anomaly detection (Chinese medicinal herbs)		Precision	AUC
	6	0.75±0	0.8333±0
	12	1±0	1±0
	24	1±0	1±0
	36	1±0	1±0

Table 11 presents the impact of training set size on classification, regression, and anomaly detection tasks. For Milk classification, accuracy increases sharply with larger datasets. This steep gain highlights the critical role of sufficient data for learning reliable class boundaries in spectral classification.

For Corn regression, performance also improves with additional training samples. These results suggest that regression benefits strongly from larger datasets, as more samples enable better approximation of continuous relationships.

In contrast, Chinese medicinal herbs anomaly detection is largely insensitive to labeled data size: even with only six training samples (3 normal samples, 2 inter class anomalies and 1 intra class anomalies), it already delivers strong performance with high Precision and AUC, and the metrics quickly approach saturation as more data are added. This robustness likely reflects pronounced separability between normal and anomalous spectra in the learned feature space, enabling a reliable decision rule from only a handful of examples.

Overall, these results reveal task dependent sensitivity to training set size: classification and regression performance depend heavily on dataset scale, while anomaly detection appears inherently robust and stable regardless of training size.

图 6. 分类与回归任务的相对平均偏差（RAD）温度消融分析。

为了进一步验证温度的影响，我们在图 6 中额外绘制了相对平均偏差（RAD）曲线。对于分类任务，RAD 在 T = 0.3 时达到最小值，在 T = 0.7 时达到最大值，表明中等温度能稳定模型的预测，而较高温度则导致不稳定的偏差。对于回归任务，RAD 在 T = 0.1 时最低，在 T = 0.9 时最高，进一步证实回归任务需要较低的温度以最小化预测方差。综上所述，RAD 分析定量地确认：分类任务受益于适度的随机性，回归任务倾向于低温，而异常检测则无论温度如何均保持稳定。

综上所述，这些发现表明最佳温度设置取决于任务类型：分类任务受益于中等温度值，回归任务倾向于较低温度值，而异常检测任务在不同温度下均保持鲁棒性。

3.5.2 数据集规模

在实验中保持测试集大小固定，改变训练样本数量揭示了三种任务不同的数据效率特征。表 11 展示了不同训练集规模对分类、回归和异常检测模型性能的影响。

表 11 不同数据集规模的实验结果

Task	训练集规模	指标	
分类（牛奶）		准确率 (%)	
	72	30.83±3.85	
	144	47.50±3.00	
	216	96.10±1.52	
回归（玉米）		R2	RMSE
	16	0.8354±0.0236	0.2114±0.0152
	32	0.8472±0.0281	0.2035±0.0184
	48	0.9252±0.0065	0.1427±0.0063
异常检测（中药材）		精度	AUC
	6	0.75±0	0.8333±0
	12	1±0	1±0
	24	1±0	1±0
	36	1±0	1±0

表 11 展示了训练集规模对分类、回归和异常检测任务的影响。对于牛奶分类，准确率随数据集增大而急剧提升。这一显著增长凸显了充足数据在学习光谱分类中可靠类别边界时的关键作用。

对于玉米回归任务，性能也随着训练样本的增加而提升。这些结果表明，回归任务极大地受益于更大的数据集，因为更多的样本能够更好地近似连续关系。

相比之下，中药材异常检测对标注数据规模 largely 不敏感：即使仅使用六个训练样本（3 个正常样本、2 个类间异常和 1 个类内异常），它已能凭借高精确率和 AUC 展现出强劲的性能，且随着数据量的增加，各项指标迅速趋于饱和。这种鲁棒性可能反映了在学习到的特征空间中，正常光谱与异常光谱之间具有显著的分离性，从而使得仅凭少量样本即可建立可靠的决策规则。

总体而言，这些结果揭示了对训练集规模的任务依赖性敏感度：分类和回归性能高度依赖于数据集规模，而异常检测则表现出固有的鲁棒性和稳定性，不受训练规模的影响。

4. Conclusion

We have presented an LLM-Driven Unified Agent framework for multi-task Reasoning in Infrared Spectroscopy that seamlessly combines literature guided preprocessing, feature extraction, and multi-task reasoning. By harnessing few-shot prompting, the system attains robust performance on classification, regression, and anomaly detection across heterogeneous spectral datasets, outperforming or matching traditional machine learning and simple deep learning baselines with minimal training data. Ablation studies underscore the roles of LLM sampling temperature, dataset size, and model selection in optimizing inference quality.

Future work will extend this agent architecture to accommodate image based spectral modalities, incorporating modules for multispectral and hyperspectral imaging. By integrating spatial-spectral feature extraction and domain specific LLM prompting, the enhanced framework will address complex analytical scenarios, such as material mapping and remote sensing, further broadening the applicability of intelligent, automated spectral analysis.

CRediT authorship contribution statement

Zujie Xie: Methodology, Data curation, Validation, Formal analysis, Visualization, Software, Writing. **Zixuan Chen:** Data curation, Validation, Formal analysis, Visualization, Writing. **Jiheng Liang:** Conceptualization, Methodology. **Xiangyang Yu:** Conceptualization, Supervision, Writing. **Ziru Yu:** Conceptualization, Methodology, Supervision, Writing.

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4. 结论

我们提出了一种用于红外光谱多任务推理的 LLM 驱动统一智能体框架，该框架无缝融合了文献引导的预处理、特征提取和多任务推理。通过利用少样本提示，该系统在异构光谱数据集的分类、回归和异常检测任务中实现了稳健的性能，仅需极少的训练数据即可超越或媲美传统机器学习和简单深度学习基线。消融研究强调了 LLM 采样温度、数据集规模和模型选择在优化推理质量中的作用。

未来的工作将扩展此代理架构，以容纳基于图像的光谱模态，并纳入多光谱和高光谱成像模块。通过整合空谱特征提取与领域特定的大语言模型提示，增强后的框架将应对材料 mapping 和遥感等复杂分析场景，进一步拓宽智能自动化光谱分析的适用范围。

CRediT 作者贡献声明

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